The maximum number of cycles in a graph of large rank

The formulas of Horak and Kim are exact for every large rank

KNT · Astra · Fable 5.2

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Let $\psi(r)$ be the largest number of cycles in a graph of cyclomatic number r. We prove that, for all sufficiently large r,

$$\psi(2q+1) = 2^{2q} + \left(2q + \tfrac{1}{2}\right)2^q - 3q, \qquad \psi(2q+2) = 2^{2q+1} + \left(2q + \tfrac{9}{2}\right)2^q - 3q - 4.$$

Cubic graphs of Horak and Kim have these numbers of cycles. In particular $\psi(r) \sim 2^{r-1}$, which Entringer and Slater conjectured in 1981. Astra and Fable 5.2 are language models. Section 1.4 states the part of each author. The theorem has a formal proof in Lean 4, and the Lean kernel has checked it.

The proof in seven sentences

1. Subdividing an edge keeps the number of cycles, and splitting a vertex of degree four or more does not lower it. So some cubic multigraph G of rank r has $\psi(r)$ cycles.
2. Fix a short cycle W and an edge e of W. For a cycle C through e, the edges of C without an end on W form a forest, and the same holds for $C + W$. By the *forest estimate*, at most $\frac{1}{2} + o(1)$ of all even subgraphs have this property, so $\psi(r) \le (\frac{1}{2} + o(1))2^r$.
3. A cut with at most four edges divides G into two parts, and a cycle across the cut consists of paths in the parts. A part with more than half of the possible paths between two ends, by a fixed margin, is a ladder up to a bounded error, and it has few cycles.
4. So a graph with such a cut is a ring of copies of $K_{2,2}$ and of single edges, or it has at most $2^{r-1} + O(2^{r/2})$ cycles.
5. A ring has at most $H(r)$ cycles, by a direct count; $H(r)$ is the right side of the formulas above.
6. Without such a cut, G has girth at least five, and it has fewer than 2^{r-1} cycles: by a second-moment argument, G contains two large pieces with small boundaries, and each piece has a forest proportion of at most $\frac{3}{5}$.
7. $H(r) - 2^{r-1}$ is at least $(r-2)2^{r/2-1}$. So $\psi(r) = H(r)$ for large r.

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1 The result and the plan

1.1 The result

Throughout, $\log$ is the binary logarithm, and $\log^{(j)}$ is its j-fold iterate. For a graph G, an *even subgraph* is a set of edges that meets every vertex an even number of times. With symmetric difference as addition, the even subgraphs form a vector space $\mathcal{Z}(G)$ over $\mathbb{F}_2$. Its dimension is

$$\beta(G) = |E(G)| - |V(G)| + \mathrm{comp}(G),$$

where $\mathrm{comp}(G)$ is the number of components. We call $\beta(G)$ the *rank* of G; other names are cyclomatic number and cycle rank. A *cycle* is a connected 2-regular subgraph. We identify a subgraph without isolated vertices with its edge set. Then a cycle is a nonzero element of $\mathcal{Z}(G)$, so a graph of rank r has at most $2^r - 1$ cycles.

Let $\psi(r)$ be the largest number of cycles in a graph of rank r.

- Entringer and Slater [4] proved that $2^{r-1} < \psi(r) < 2^r$ for $r \geq 3$, and they conjectured that $\psi(r) \sim 2^{r-1}$.
- Aldred and Thomassen [2] proved that $\psi(r) \leq \frac{15}{16}2^r$.
- Alahmadi, Aldred, dela Cruz, Solé and Thomassen [1] proved that a connected graph with p vertices and more than $2p + 3\log(3p)$ edges has at most 2^{r-1} cycles.
- Horak and Kim [5] found cubic graphs of rank r with $H(r)$ cycles, where

$$H(2q+1) = 2^{2q} + \left(2q + \tfrac{1}{2}\right)2^q - 3q, \qquad H(2q+2) = 2^{2q+1} + \left(2q + \tfrac{9}{2}\right)2^q - 3q - 4. \tag{1.1}$$

For odd rank $r = 2q + 1$ the graph of Horak and Kim is a ring of q copies of $K_{2,2}$, each joined to the next by two edges. For even rank, one more edge is inserted across the ring (Figure 1). We call this graph R_r.

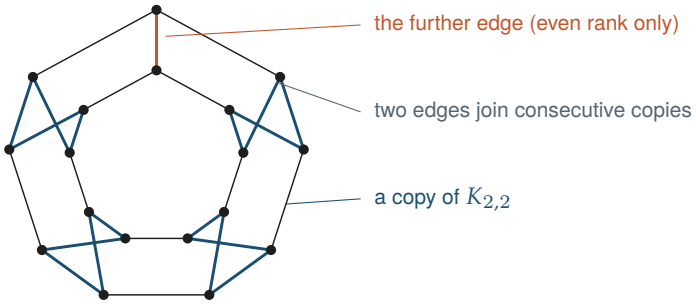


Figure 1. The graph R_{10}: a ring of four copies of $K_{2,2}$ and one further edge. It has rank 10 and $H(10) = 696$ cycles.

Theorem 1.1 (the maximum number of cycles)

There is an integer R such that $\psi(r) = H(r)$ for every $r \geq R$.

In particular $\psi(r) = 2^{r-1} + (1 + o(1))\, r\, 2^{\lfloor (r-1)/2 \rfloor}$, so $\psi(r) \sim 2^{r-1}$. The proof gives no value for R.

> **Fine print.** The theorem is false for small r. For example, take the vertices u_i, v_i with $i \in \mathbb{Z}/26$ and the edges u_iu_{i+1}, u_iv_i and v_iv_{i+10}. This generalised Petersen graph has rank 27, and a direct count by computer gives 67 959 973 cycles, while $H(27) = 67\,325\,913$.

1.2 The four cases of the proof

The number of cycles does not change when we delete a vertex of degree at most one, or replace a vertex of degree two and its two edges by one edge. A vertex of degree at least four can be split into vertices of degree three, and no cycle is lost. So it suffices to count the cycles of a connected multigraph G of rank r in which all vertices have degree three (Lemma 3.6). The graphs R_r show that $\psi(r) \geq H(r)$.

For the upper bound there are four cases (Figure 2). A *cyclic cut* is a set of edges whose deletion leaves two parts that each contain a cycle. The number $H(r) - 2^{r-1}$ is of order $r\,2^{r/2}$, so the four bounds give the theorem (Section 8).

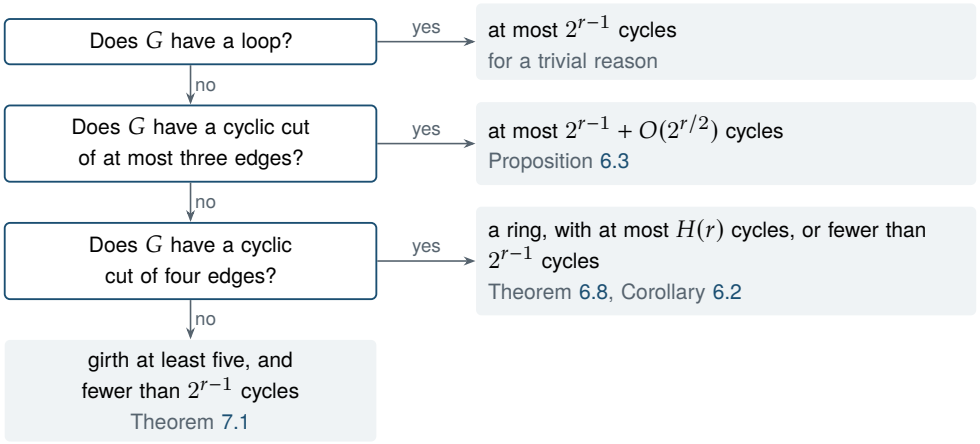


Figure 2. The four cases for a connected cubic multigraph G of large rank r.

1.3 Outside parts and the forest estimate

Fix a set P of vertices of G. For an even subgraph ξ we call the set of its edges without an end in P its *outside part* (Figure 3). If a cycle meets P, then its outside part is a proper subset of a cycle, so it is a forest.

> **IDEA | cycles and outside parts**
>
> Let W be a cycle, e an edge of W, and $P = V(W)$. For every cycle C through e, the even subgraphs C and $C + W$ have the same outside part, and it is a forest. By the *forest estimate* (Theorem 3.1), the outside part is a forest for at most $\frac{1}{2} + o(1)$ of all even subgraphs when P is small. An edge recursion then gives $\psi(r) \leq (\frac{1}{2} + o(1))2^r$ (Corollary 3.7).

In the cases 2 and 3 a cycle consists of paths in the parts of G. By the forest estimate, a part with more than half of the possible paths between two of its ends, by a fixed margin, is a ladder up to a bounded error, and it has few cycles (Section 5). In the case 3 we also close a part by a small tree, a *cap*, and apply the bound $(\frac{1}{2} + o(1))2^r$ to the closed part (Lemma 6.5). In the case 4 the forest estimate is not sufficient, because we need a bound below $\frac{1}{2}$ and its error term is positive.

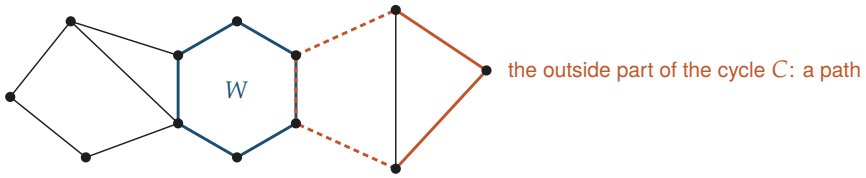


Figure 3. A cycle W and a cycle C through an edge of W. The edges of C with an end on W are dashed. The other edges of C form its outside part for $P = V(W)$, which is a forest. An even subgraph whose outside part contains a cycle is not a cycle through W.

In this case a second-moment argument gives two disjoint pieces of G with small boundaries, and each piece contributes a factor of at most $\frac{3}{5}$ to the proportion of forests (Section 7).

1.4 Authorship and verification

KNT is a person. Astra and Fable 5.2 are language models: GPT-6 Astra and Claude Fable 5.2. KNT directed the work at each step. The forest estimate and its proof are from [6]; KNT obtained them in September 2026 with Astra, for a problem of Erdős on pancyclic graphs (see also [8]). In November 2026 the two models studied the number of cycles in parallel, in one shared set of files.

The parts have these origins. Fable 5.2 found the edge recursion, the localisation and layer argument, the chain of nested cycles, the parallel-link lemma and Lemma 6.7. Astra found the path structure of Section 5, the gap theorem, the first analysis of cuts, the count for rings, and the first proof for graphs without a small cyclic cut: an argument for bounded girth, and a packing bound and two girth scales for large girth. In its first form, Section 7 gave the bound $\frac{1}{2}$ for the proportion of forests, and the proof of Theorem 1.1 then used an induction on the rank. Astra improved the bound to a constant below $\frac{1}{2}$, and the induction is no longer necessary. Fable 5.2 shortened the arguments for paths and cuts and wrote this text. After the formal proof was complete, Astra found the cap of Lemma 6.5, which removed two cases from the proof of Theorem 6.8, and the marked set of Case 4 in Section 8. Fable 5.2 then found that the chain of Proposition 7.9 also gives the packing bound, and that Proposition 3.5 serves Sections 3 and 7, that one girth scale is sufficient (Proposition 7.11), and that a large piece of girth five has few forests (Lemma 7.8). So the argument for bounded girth is not necessary.

Theorem 1.1 has a formal proof in Lean 4. The formal statement is `Erdos1016.ExactMaximum.exactMaximum_caps`: there is a number R such that `maxCyclesr=horakKimr` for every $r \geq R$. Here `maxCyclesr` is the largest number of cycles of a simple graph with cyclomatic number r on a vertex set $\{0, \dots, n-1\}$, a cycle is counted by its edge set, and `horakKimr` is $H(r)$. A reader who trusts the Lean kernel needs to read these definitions only. The table lists the formal statements of the main steps. All of them are proved.

Statement	Declaration
Theorem 3.1	`Erdos1016.ShortProof.eventually_marked_forest_bound_logSquared`
Corollary 3.7	`Erdos1016.cycleCountTheorem`; with parallel edges: `EM.numCycles_le_of_large_rank_margin`
Lemma 3.6, as the upper bound uses it	`EMS.CubicReduction.cubicReductionStatement`
Equation (6.1) and the count of Lemma 6.4	`Net.numCycles_of_three_crossing`, `Net.numCycles_of_four_crossing`
Lower bound of Corollary 6.2: for every $r \geq 3$ a graph of rank r with at least $H(r)$ cycles	`EM.lowerBoundStatement`
Case 1 of Section 8 (a loop)	`EM.numCycles_le_horakKim_of_loop`
Theorem 5.4 for $\varepsilon = \frac{1}{1000}$, with Lemma 5.2 and Proposition 5.3; Lemma 5.5	`EM.fewCyclesStatement`, `EM.gadget_third_pair`, `EM.gadget_cross_pairs`, `EM.gadget_opposite_pair`
Case 2: Proposition 6.3 with some constant	`EM.threeCutStatement`
Theorem 5.8	`EMS.gapStatement`
Lemma 6.5	`EMS.Shortening.TreeCapCycleBound.exists_thousandth_bound`
Lemma 6.7; Lemma 6.6 and the cases of Theorem 6.8, as arithmetic	`Net.two_mul_numPathPairs_le_of_full`, `EMS.Shortening.FourCutTreeCapArithmetic.common_full_cross_of_tree_caps`
In Theorem 6.8: the sides have no bridge; four full pairs in both sides give at most $H(r)$ cycles	`EMS.Shortening.MinimalCutBridgeless.four_cut_shore_bridgeless`, `EMS.ringCaseStatement`
Case 3: Theorem 6.8 with the rings	`EM.fourCutStatement_caps`
Case 4 for a graph with a cycle shorter than a number D_*	`EM.shortCycleStatement`, with `EMS.cyclicShoreStatement`
Case 4 for girth at least D_*, with a packing bound $150\,d^3(s+1)2^d$ outside a deleted set of s vertices	`EMS.Shortening.ChainPacking.exists_cutoff`, `EMS.Shortening.ChainForestProbability.exists_cutoff`, `EM.largeGirthStatement_chain`
Upper bound: at most $H(r)$ cycles for large r	`EM.upperBoundStatement_caps`
Theorem 1.1; equation (1.1) for both parities	`EM.exactMaximum_caps`; `EM.maxCycles_formula_caps`

Short names: `EM` is `Erdos1016.ExactMaximum`, `EMS` is `Erdos1016.ExactMaximumSupport`, and `Net` is `Erdos1016.BoundaryTrace.Network`.

The files are in [9], on the branch `feat/cycle-count-bound`. Commit `3a80aa9` has a first formal proof, `EM.exactMaximum`. It follows the first form of this text, which had no cap, marked a shortest cycle in the last case, and had another proof of the packing bound. The table names the second formal proof, which follows this text; at the date of this draft its files are not in a public commit. The files contain no `sorry`: Fable 5.2 wrote `Erdos1016/ExactMaximum/`, and Astra wrote `Erdos1016/ExactMaximumSupport/` and the formal proof of Theorem 3.1. Astra's files contain the formal proofs of Section 5.3, of the upper bound for rings, of Lemma 6.5 with the arithmetic of Theorem 6.8, and of Section 7. The Lean kernel accepts the formal proofs with the axioms `propext`, `Classical.choice` and `Quot.sound` only.

The kernel check covers the formal statements, and no machine has checked the text of this paper. A formal proof can differ from the text. In four places the method is different. The formal proof of Theorem 3.1 bounds the links of two cycles, which is Step 2 of Proposition 3.5, with Lemma 2.4. The formal proof of Corollary 3.7 uses the forest estimate in its form for a simple graph, which is in [8]. The formal proofs of Lemma 5.6 and Lemma 5.7 use two disjoint paths between the ports and exact numbers of paths in a capped ladder. The largest difference is in Case 4. The formal proof has a number $D_* \geq 256$ and two cases. For a graph with a cycle shorter than D_* it sorts the cycles through an edge of a shortest cycle by the vertex where they leave it. For girth at least D_* it uses pieces of large girth, and it raises the girth in stages by the deletion of maximal families of short cycles. In other places the details differ. For example, the formal cap of Lemma 6.5 has a vertex in the middle of the edge xy, and the formal proof of Theorem 5.4 calls a thin layer ordinary if at most one of its two vertices is on a pair of parallel edges. So the text can contain errors that the formal proof does not have. The two models have read the text, and no referee has. Scripts in the repository test the statements of Sections 5 and 6 on small graphs. These tests do not replace a proof.

2 Tools

NOTATION | multigraphs, cuts, regions and outside parts

- From here on a graph may have loops and parallel edges. We say *multigraph* to stress this, and *simple graph* to exclude it. A loop adds two to the degree of its vertex. A loop is a cycle of length one, and two parallel edges form a cycle of length two.
- A multigraph is *cubic* if every vertex has degree three, and *subcubic* if every vertex has degree at most three. The *girth* is the length of a shortest cycle.
- $c(G)$ is the number of cycles of G, and $c_e(G)$ is the number of cycles through an edge e. If we replace an edge by a path, the rank and the number of cycles do not change. So $\psi(r)$ is also the largest number of cycles of a multigraph of rank r.
- For a set A of vertices, $G[A]$ is the induced subgraph, $G - A$ is $G[V(G) \setminus A]$, and δA is the set of edges with exactly one end in A.
- A *region* is a nonempty set A such that $G[A]$ is connected. A region is *cyclic* if $G[A]$ contains a cycle.
- A *cyclic cut* is a set δX such that $G[X]$ and $G - X$ each contain a cycle; its *size* is $|\delta X|$.
- Throughout, ξ is a uniformly random element of $\mathcal{Z}(G)$.
- For a set P of vertices and an even subgraph ξ, the *outside part* of ξ is $\xi \cap E(G - P)$, the set of its edges without an end in P. We write $q(G, P)$ for the probability that the outside part of ξ is a forest.

Lemma 2.1. Let $Z \geq 0$ be a random variable with $\mu = \mathbb{E}Z > 0$ and $\mathbb{E}Z^2 \leq 2\mu^2 + K\mu$. Then $\mathbb{P}(Z = 0) \leq \frac{1}{2} + K/(4\mu)$.

Proof. By the Cauchy–Schwarz inequality $\mu^2 \le \mathbb{E}Z^2 \cdot \mathbb{P}(Z>0)$, so $\mathbb{P}(Z=0) \le 1-\mu/(2\mu+K) \le \frac{1}{2} + K/(4\mu)$. ∎

Let A and B be disjoint regions of a connected multigraph G. A *link* between them is an edge with one end in A and the other in B, or a component of $G-(A\cup B)$ that is joined by an edge to A and by an edge to B (Figure 4). Let $\ell(A,B)$ be the number of links. For a region A let O_A be the event $\xi \cap \delta A = \emptyset$.

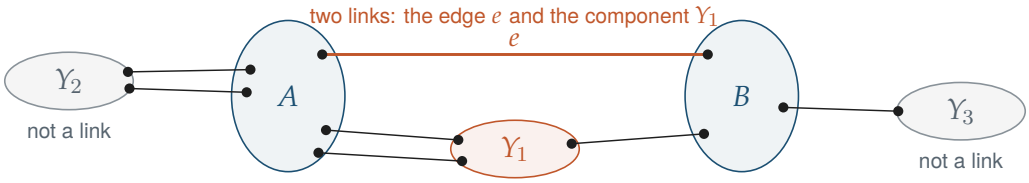


Figure 4. Links between two regions A and B. The edge e and the component Y_1 of $G-(A\cup B)$ are links. The components Y_2 and Y_3 are joined to one region only, so they are not links. Here $\ell(A,B)=2$.

Lemma 2.2 (Link formula)**.** Let A and B be disjoint regions of a connected multigraph G. Then $\mathbb{P}(O_A) = 2^{\mathrm{comp}(G-A)-|\delta A|} \ge 2^{-|\delta A|}$ and

$$\mathbb{P}(O_A \cap O_B) = 2^{\ell(A,B)-1}\,\mathbb{P}(O_A)\,\mathbb{P}(O_B).$$

Conditionally on O_A, the set $\xi \cap E(G[A])$ is uniform on $\mathcal{Z}(G[A])$. Conditionally on $O_A \cap O_B$, the sets $\xi \cap E(G[A])$ and $\xi \cap E(G[B])$ are independent and uniform on $\mathcal{Z}(G[A])$ and $\mathcal{Z}(G[B])$.

Proof. The even subgraphs that avoid δA are the sums of an even subgraph of $G[A]$ and an even subgraph of $G-A$. So $\mathbb{P}(O_A) = 2^{\beta(G[A])+\beta(G-A)-\beta(G)}$, and a count of edges and vertices shows that the exponent is $\mathrm{comp}(G-A)-|\delta A|$. In the same way

$$\mathbb{P}(O_A\cap O_B) = 2^{\beta(G[A])+\beta(G[B])+\beta(G-A-B)-\beta(G)} = 2^{\mathrm{comp}(G-A-B)+1-|\delta A|-|\delta B|+e(A,B)},$$

where $e(A,B)$ is the number of edges between A and B. Let a, b and k be the numbers of components of $G-A-B$ that are joined to A only, to B only, and to both. Then $\mathrm{comp}(G-A-B) = a+b+k$, $\mathrm{comp}(G-A) = a+1$ and $\mathrm{comp}(G-B) = b+1$. The formula follows, because $\ell(A,B) = k + e(A,B)$. The conditional laws are read off from the two decompositions as direct sums. ∎

IDEA | the correlation of two cut events

By Lemma 2.2, two cut events are independent when the regions have one link, and their joint probability is twice the product when the regions have two links. The constant $\frac{1}{2}$ of the forest estimate comes from this factor two, through Lemma 2.1.

Pairs with three or more links are rare:

Lemma 2.3 (Exceptional partners). Let A be a region of a connected multigraph G, and let $B_1, \dots, B_k$ be pairwise disjoint regions, disjoint from A, with $\ell(A, B_i) \geq 3$ for all i. Then $k \leq |\delta A|$.

Proof. Contract A to a vertex a and each B_i to a vertex b_i, and delete the loops that arise. This changes neither $d = |\delta A|$ nor the numbers $\ell(A, B_i)$, because the sets B_j with $j \neq i$ are connected and are contracted inside components of $G - (A \cup B_i)$. Let F be a spanning forest of $G - a$, and for every edge $e = au$ at a attach a new leaf τ_e to u. Call the resulting forest F^+ and its d new leaves *terminals*.

Fix i. Every link between a and b_i yields a path of F^+ from b_i to a terminal. For an edge $e = ab_i$ it is the edge $b_i\tau_e$. For a component Y of $G - \{a, b_i\}$ joined to both, choose an edge $e = au$ with $u \in Y$ and take the path of F^+ from b_i to τ_e; after its first vertex it stays in $Y \cup \{\tau_e\}$. Paths that come from different links start with different edges at b_i, and the union of any two of them is a path between two terminals. Let $F^\circ \subseteq F^+$ be the union of all paths between terminals. As $\ell(a, b_i) \geq 3$, the vertex b_i has degree at least 3 in F°. All leaves of F° are terminals, and a forest with at most d leaves has at most $d - 2$ vertices of degree at least 3. Hence $k \leq d - 2$ if $k \geq 1$. ∎

Lemma 2.4 (Many small cuts). Let $A_1, \dots, A_k$ be pairwise disjoint cyclic regions of a connected multigraph G with $|\delta A_i| \leq d$ for all i. Then

$$\mathbb{P}\big(\xi \cap E(G[A_i]) \text{ is a forest for every } i\big) \;\leq\; \frac{1}{2} + \frac{(d+1)\,2^{d-1}}{k}.$$

Proof. Let E_i be the event that O_{A_i} holds and $\xi \cap E(G[A_i]) \neq \emptyset$. On E_i the set $\xi \cap E(G[A_i])$ is a nonempty even subgraph, hence not a forest. So it suffices to bound $\mathbb{P}(Z = 0)$ for $Z = \sum_i \mathbf{1}_{E_i}$. By Lemma 2.2, $p_i := \mathbb{P}(E_i) = \mathbb{P}(O_{A_i})\big(1 - 2^{-\beta(G[A_i])}\big) \geq 2^{-d-1}$ and $\mathbb{P}(E_i \cap E_j) = 2^{\ell(A_i, A_j)-1} p_i p_j$ for $i \neq j$. If $\ell(A_i, A_j) \leq 2$, this is at most $2p_ip_j$. Otherwise we use $\mathbb{P}(E_i \cap E_j) \leq p_i$; by Lemma 2.3 this happens for at most d indices j, for each i. Hence $\mathbb{E}Z^2 \leq \mu + 2\mu^2 + d\mu$ with $\mu = \mathbb{E}Z \geq k\,2^{-d-1}$, and Lemma 2.1 gives the claim. ∎

3 The forest estimate

Theorem 3.1 (Forest estimate, KNT [6]). There is r_0 such that the following holds for all $r \geq r_0$. Let Γ be a connected subcubic multigraph of rank r. Let P be a set of at most $2r/\log^2 r$ vertices such that every vertex outside P has degree three and $\Gamma[P]$ has at most $(\log r)/100$ components. Then

$$q(\Gamma, P) \;\leq\; \frac{1}{2} + \frac{20}{\log^{(3)} r}.$$

This paper uses the theorem only when $\Gamma[P]$ is connected. The form with several components has the same proof, and [8] needs it: there the theorem gives an estimate for a union of cycles in a simple graph, which is the form of [6]. The proof is that of [6].

IDEA | the proof in three steps

1. If $\Gamma - P$ has many disjoint short cycles, the estimate follows from Lemma 2.4. So we may delete a small maximal family of them, and a graph J of large girth remains.
2. The graph J has many cycles with between $6 \log r$ and $\log r \sqrt{\log \log r}$ edges. Their total weight $\sum_C 2^{-|C|}$ is at least $(\frac{1}{8} - o(1)) \ln \log \log r$ (Lemma 3.4).
3. For such a cycle C, the event that C is a component of ξ has probability at least $2^{-|C|}$, and then the outside part of ξ is not a forest. Most pairs of these events are correlated by a factor of at most two (Lemma 2.2).

With the factor two of step 3, the second moment of the number Z of these events is at most $2\mu^2 + K\mu$, and Lemma 2.1 then gives $\mathbb{P}(Z = 0) \le \frac{1}{2} + K/(4\mu)$.

Proposition 3.5 has the steps 2 and 3 in a form that Section 7 uses again.

3.1 Cycles of moderate length in a graph of large girth

In this subsection Γ is a connected subcubic multigraph, and U is a set of its vertices such that every vertex outside U has degree 3. We suppose that $J = \Gamma - U$ is a simple graph of girth greater than D, where $D \ge 3$ is an integer, and we put $s = \lfloor (D-1)/2 \rfloor$. Let $L \ge 1$ be an integer, and let $\mathcal{C}$ be the set of the cycles of J of length at most L. Let v be the number of vertices of J, and let

$$d_J = \sum_{u \in V(J)} \bigl(3 - \deg_J u\bigr)$$

be its *deficit*. This is the number of edges between $V(J)$ and U, so $d_J \le 3|U|$.

A walk $u_0 u_1 \cdots u_\ell$ is *non-backtracking* if $u_{i+1} \ne u_{i-1}$ for $0 < i < \ell$. Let $N_\ell(u, w)$ be the number of non-backtracking walks of length ℓ from u to w in J.

Lemma 3.2. If $\ell > s$, then $N_\ell(u, w) \le 3 \cdot 2^{\ell - s - 1}$ for all u, w. Consequently, for every vertex w of J,

$$\sum_{C \in \mathcal{C},\ C \ni w} 2^{-|C|} \ \le\ L\, 2^{-s}. \tag{3.1}$$

Proof. There are at most $3 \cdot 2^{\ell - s - 1}$ ways to choose the first $\ell - s$ steps. The last s steps form a non-backtracking walk of length s between two given vertices. Such a walk is a path, because s is less than the girth. There is at most one such path, because two of them would give a cycle of length at most $2s < D$. For the second statement, a cycle of length ℓ through w can be traversed from w in two directions. So there are at most $\frac{1}{2} N_\ell(w, w)$ of them, and $\ell > D > s$. Hence the cycles of each length $\ell \le L$ contribute at most $\frac{3}{4}\, 2^{-s}$. ∎

Let B be the non-backtracking matrix of J. Its rows and columns are indexed by the directed edges $(i \to j)$ of J, two for each edge. The entry in row $(i \to j)$ and column $(j' \to k)$ is 1 if $j' = j$ and $k \neq i$, and it is 0 otherwise.

Lemma 3.3. Suppose that $d_J < v$. Then, for every even $\ell \geq 2$,

$$\operatorname{tr} B^\ell \;\geq\; \bigl(2 - d_J/v\bigr)^\ell - 3v\,2^{\ell/2}.$$

Proof. Let A be the adjacency matrix of J and Δ the diagonal matrix of its degrees, and put $M(\lambda) = \lambda^2 I - \lambda A + \Delta - I$.

(1) *If $M(\lambda)g = 0$ for a real $\lambda > 1$ and a vector $g \neq 0$, then λ is an eigenvalue of B.* A direct computation shows that $f(i \to j) = \lambda g(j) - g(i)$ satisfies $Bf = \lambda f$. And $f = 0$ would give $g(i) = \lambda g(j) = \lambda^2 g(i)$ along every edge, hence $g = 0$; at an isolated vertex, the equation $M(\lambda)g = 0$ is $(\lambda^2 - 1)g = 0$.

(2) *If $Bf = \lambda f$ with $f \neq 0$ and $\lambda^2 \neq 1$, then $M(\lambda)g = 0$ for $g(j) = \sum_{k \sim j} f(j \to k)$, and $g \neq 0$.* The eigenvalue equation at $(i \to j)$ and at $(j \to i)$ reads $\lambda f(i \to j) = g(j) - f(j \to i)$ and $\lambda f(j \to i) = g(i) - f(i \to j)$. Eliminating $f(j \to i)$ gives $(\lambda^2 - 1) f(i \to j) = \lambda g(j) - g(i)$. This shows that $g \neq 0$. Summing it over the neighbours j of i gives $M(\lambda)g = 0$.

(3) *B has a real eigenvalue $\lambda_1 \geq 2 - d_J/v$.* The matrix $M(2) = 2(\Delta - A) + (3I - \Delta)$ is positive semidefinite, while $\mathbf{1}^\top M(\lambda)\mathbf{1} = (\lambda - 1)(v\lambda - 2v + d_J)$ vanishes at $\lambda^* = 2 - d_J/v > 1$. The smallest eigenvalue of the symmetric matrix $M(\lambda)$ is continuous in λ, at most 0 at λ^* and at least 0 at 2. So $M(\lambda_1)$ is singular for some $\lambda_1 \in [\lambda^*, 2]$, and (1) applies.

(4) *Every non-real eigenvalue λ of B satisfies $|\lambda| \leq \sqrt{2}$.* With g as in (2), $g^* M(\lambda) g = 0$ is a quadratic equation $a\lambda^2 - b\lambda + c' = 0$ with real coefficients $a = g^* g > 0$, $b = g^* A g$ and $c' = g^*(\Delta - I)g \leq 2a$. Its roots are λ and $\bar\lambda$, so $|\lambda|^2 = c'/a \leq 2$.

Now $\operatorname{tr} B^\ell$ is the sum of the ℓth powers of the $2|E(J)| \leq 3v$ eigenvalues of B. For even ℓ the real eigenvalues contribute nonnegative terms, one of which is at least $(2 - d_J/v)^\ell$, and the non-real ones contribute at least $-3v\,2^{\ell/2}$ in total. ∎

Facts (2) and (4) are part of the Ihara–Bass theory [3, 7]. The test vector $\mathbf{1}$ in (3) is from [6]. With this test vector the argument needs no assumption of regularity or expansion.

Lemma 3.4 (Cycle weight)**.** Suppose that $d_J < v$, and let $\ell_0 \leq L$ be a positive integer. The total weight $\mu := \sum_{C \in \mathcal{C}} 2^{-|C|}$ satisfies

$$\Bigl(1 + \tfrac{3}{2} L^2 2^{-s}\Bigr)\mu \;\geq\; \Bigl(1 - \frac{L\,d_J}{2v} - 3v\,2^{-\ell_0/2}\Bigr) \sum_{\ell} \frac{1}{2\ell},$$

where the sum is over the even ℓ with $\ell_0 \leq \ell \leq L$.

Proof. $\operatorname{tr} B^\ell$ is the number of closed walks $u_0u_1\cdots u_\ell = u_0$ in J with $u_{i+1} \neq u_{i-1}$ for all i modulo ℓ. A cycle of length ℓ gives 2ℓ of them; let n_ℓ be the number of the others. Then

$$T := \sum_{\ell\le L} \frac{\operatorname{tr} B^\ell}{2\ell\, 2^\ell} = \mu + \sum_{\ell\le L} \frac{n_\ell}{2\ell\, 2^\ell}.$$

Consider a walk counted by n_ℓ, and let j be the least index with $u_j \in \{u_0,\dots,u_{j-1}\}$, say $u_j = u_i$. Then $j < \ell$, the walk $C = u_iu_{i+1}\cdots u_j$ is a cycle, of length q say, and $u_ju_{j+1}\cdots u_\ell u_1\cdots u_i$ is a non-backtracking closed walk of length $\ell - q > 0$ from u_i to itself. It contains a cycle, so $\ell - q > D > s$. The original walk is determined by C, by the vertex u_i and a direction on C ($2q$ choices), by the complementary walk (at most $3\cdot 2^{\ell-q-s-1}$ choices, by Lemma 3.2), and by the position of u_0 on the latter (at most ℓ choices). Hence

$$\sum_{\ell\le L} \frac{n_\ell}{2\ell\, 2^\ell} \;\le\; \sum_{\ell\le L}\sum_{C\in\mathcal{C}} \tfrac{3}{2}\,|C|\,2^{-|C|-s} \;\le\; \tfrac{3}{2}\,L^2\,2^{-s}\,\mu,$$

and $T \le (1 + \frac{3}{2}L^2 2^{-s})\mu$. On the other hand, all terms of T are nonnegative. For even ℓ with $\ell_0 \le \ell \le L$, Lemma 3.3 gives

$$2^{-\ell}\operatorname{tr} B^\ell \;\ge\; \Big(1 - \frac{d_J}{2v}\Big)^\ell - 3v\,2^{-\ell/2} \;\ge\; 1 - \frac{L\,d_J}{2v} - 3v\,2^{-\ell_0/2}. \qquad \blacksquare$$

The second moment. Let ξ be uniform on $\mathcal{Z}(\Gamma)$. For $C \in \mathcal{C}$ let I_C be the event that C is a component of ξ. As the vertices of C have degree 3 in Γ, this is the event $E(C) \subseteq \xi$. It is also the event that $\mathcal{O}_{V(C)}$ holds and $\xi \cap E(\Gamma[V(C)]) = E(C)$. Put

$$a_C = 2^{-|C|}, \qquad p_C = \mathbb{P}(I_C), \qquad X_C = \frac{a_C}{p_C}\,\mathbf{1}_{I_C}, \qquad Z = \sum_{C\in\mathcal{C}} X_C.$$

Because of the factor a_C/p_C, the mean of X_C is the combinatorial weight a_C for every value of p_C. The event I_C is given by $|C|$ linear conditions on $\xi \in \mathcal{Z}(\Gamma)$, which are satisfied by $\xi = C$. So $p_C \ge a_C$. Hence $0 \le X_C \le 1$, and $\mathbb{E}Z = \mu$. If $Z \neq 0$, then ξ contains a cycle of J.

Let $C \neq C'$ be in $\mathcal{C}$. If they share a vertex, then $I_C \cap I_{C'} = \emptyset$. If they are disjoint, Lemma 2.2, applied to the regions $V(C)$ and $V(C')$ of Γ, gives $\mathbb{P}(I_C \cap I_{C'}) = 2^{\ell(C,C')-1}p_Cp_{C'}$ and therefore

$$\mathbb{E}(X_CX_{C'}) = a_Ca_{C'}\,2^{\ell(C,C')-1}, \tag{3.2}$$

where $\ell(C,C')$ is the number of links between $V(C)$ and $V(C')$ in Γ. Let W_2 be the sum of $a_Ca_{C'}$ over the ordered pairs of disjoint cycles of $\mathcal{C}$ with exactly two links. Then $W_2 \le \mu^2$.

Proposition 3.5 (Logarithmic girth). There is r_2 such that the following holds for all $r \geq r_2$, with $x = \log r$ and $y = \log x$. Let Γ and P be as in Theorem 3.1, with $q(\Gamma, P) \geq \frac{2}{5}$. Let D and L be integers with $x/16 \leq D \leq x/8$ and $7x \leq L \leq x\sqrt{y}$. Let S be the union of the vertex sets of a maximal family of pairwise disjoint cycles of length at most D in $\Gamma - P$, and suppose that $|S| \leq r/x^2$. Then Γ and $U = P \cup S$ are as in this subsection, $s \geq x/32 - 1$, and

$$L^2\, 2^{-s} \leq r^{-1/60}, \qquad \mu \;\geq\; \frac{1}{4}\ln\frac{L}{6x} - \frac{1}{\sqrt{x}}, \qquad \mathbb{E}Z^2 \;\leq\; \mu^2 + W_2 + \bigl(1 + r^{-1/60}\bigr)\mu.$$

Proof. All estimates below hold for large r, uniformly in the other data. The family contains the loops and the pairs of parallel edges of $\Gamma - P$, because these are cycles of length at most two. By its maximality, $J = \Gamma - U$ is a simple graph of girth greater than D. We have $s \geq (D-2)/2 \geq x/32 - 1$, so $L^2 2^{-s} \leq 2x^2 y\, r^{-1/32} \leq r^{-1/60}$. Recall that $|P| \leq 2r/x^2$ and that the number c of components of $\Gamma[P]$ is at most $x/100$.

Step 1: components with a cycle. Let A and B be disjoint regions of Γ, and let $t \geq 0$ be an integer. We claim that at most $5 \cdot 2^t$ components Y of $\Gamma - (A \cup B)$ have all of the following properties: Y has an edge to A and an edge to B; Y avoids P; Y contains a cycle; $|\delta Y| \leq t$. Let $Y_0, Y_1, \dots, Y_n$ be such components. For $1 \leq i < j \leq n$ the graph $\Gamma - Y_i - Y_j$ is connected. Indeed, it contains the connected set $A \cup B \cup Y_0$, and every other component of $\Gamma - (A \cup B)$ is connected and has an edge to $A \cup B$. No edge joins Y_i and Y_j. So the regions Y_i and Y_j have exactly one link. Let E_i be the event that O_{Y_i} holds and $\xi \cap E(\Gamma[Y_i])$ is not empty. By Lemma 2.2, the events E_i and E_j are independent, and $\mathbb{P}(E_i) \geq 2^{-t} \cdot \frac{1}{2}$. If $n = 0$, the bound holds. Otherwise let N be the number of the events $E_1, \dots, E_n$ that occur, and put $\nu = \mathbb{E}N \geq n\, 2^{-t-1} > 0$. Then $\mathbb{E}N^2 \leq \nu^2 + \nu$, and the Cauchy–Schwarz inequality gives $\mathbb{P}(N = 0) \leq 1 - \nu^2/\mathbb{E}N^2 \leq 1/(1+\nu)$. If $\xi \cap E(\Gamma - P)$ is a forest, then no E_i occurs, because Y_i avoids P and a nonempty even subgraph is not a forest. So $\frac{2}{5} \leq q(\Gamma, P) \leq 1/(1+\nu)$. Hence $\nu \leq \frac{3}{2}$ and $n \leq 3 \cdot 2^t$, and there are at most $3 \cdot 2^t + 1$ such components.

Step 2: the links of two cycles. Put $t = \lfloor y/2 \rfloor$; then $1 \leq t \leq D/4$. Let $C, C' \in \mathcal{C}$ be disjoint, and put $U' = V(C) \cup V(C')$. Every vertex of U' lies on exactly one edge outside its own cycle, so at most $2L$ edges leave U'. At most c components of $\Gamma - U'$ meet P, because $P \cap U' = \emptyset$ and so each component of $\Gamma[P]$ lies in one of them. At most $2L/(t+1)$ components have more than t edges to U'. Of the other components that are joined to both cycles, at most $5 \cdot 2^t$ contain a cycle, by Step 1 with $A = V(C)$ and $B = V(C')$. The remaining components that are joined to both cycles induce trees, avoid P and have at most t edges to U'. They also avoid S: each cycle of the family is disjoint from U', so it lies in one component of $\Gamma - U'$, and this component contains a cycle.

Let Y be one of the remaining components, with $d_Y \leq t$ edges to U'. All its vertices have degree 3, so $3|Y| = 2(|Y|-1) + d_Y$, that is, $|Y| = d_Y - 2$. So there is a path from C to C' through Y with at most t edges. For a link that is an edge, take this edge as the path. These paths lie in J. They are pairwise disjoint, because every vertex of U' lies on exactly one edge outside its own cycle. Divide each of C and C' into arcs of at most $\lfloor D/8 \rfloor$ consecutive vertices. At most $17L/D$ arcs are necessary for each cycle, because $\lfloor D/8 \rfloor \geq D/16$ and $L \geq D$. If two of

the paths had their ends in the same arc of C and in the same arc of C', they would form, together with parts of these two arcs, a cycle of J of length at most $2t + 2\lfloor D/8 \rfloor < D$. So there are at most $(17L/D)^2$ such paths. Together,

$$\ell(C, C') \;\le\; b := c + \frac{2L}{t+1} + 5 \cdot 2^t + 289\Big(\frac{L}{D}\Big)^2 \;\le\; \frac{x}{100} + \frac{4x}{\sqrt{y}} + 5\sqrt{x} + 289 \cdot 256\, y \;\le\; \frac{x}{90}.$$

Step 3: the second moment. Call a disjoint pair $C, C' \in \mathcal{C}$ *exceptional* if $\ell(C, C') \ge 3$. Fix C, and let $\mathcal{E}$ be a maximal family of pairwise disjoint exceptional partners of C. By Lemma 2.3, $|\mathcal{E}| \le |\delta V(C)| \le L$. So the union of their vertex sets has at most L^2 vertices, and every exceptional partner of C meets this union. By (3.1), the sum of $a_{C'}$ over the exceptional partners C' of C is at most $L^3 2^{-s}$. With (3.2) and Step 2,

$$\sum_{C' \text{ exceptional}} \mathbb{E}(X_C X_{C'}) \;\le\; a_C\, 2^b L^3\, 2^{-s} \;\le\; a_C\, 2^{x/90 - x/32 + 1}\big(x\sqrt{y}\big)^3 \;\le\; a_C\, r^{-1/60}.$$

By (3.2), a disjoint pair with at most one link has $\mathbb{E}(X_C X_{C'}) \le a_C a_{C'}$, and a pair with two links has $\mathbb{E}(X_C X_{C'}) = 2a_C a_{C'}$. Intersecting pairs contribute nothing, and $\mathbb{E}X_C^2 \le \mathbb{E}X_C = a_C$. The sum over all ordered pairs is the bound for $\mathbb{E}Z^2$.

Step 4: the weight. A count of edge ends gives $3|V(\Gamma)| - 3|P| \le 2|E(\Gamma)| \le 3|V(\Gamma)|$, and $|E(\Gamma)| = |V(\Gamma)| + r - 1$. So $2r - 2 \le |V(\Gamma)| \le 2r - 2 + 3|P|$. Hence the number v of vertices of J satisfies $r \le v \le 3r$, and $d_J \le 3(|P| + |S|) \le 9r/x^2$. By Lemma 3.4 with $\ell_0 = \lceil 6x \rceil$,

$$\big(1 + \tfrac{3}{2}\, r^{-1/60}\big)\,\mu \;\ge\; \Big(1 - \frac{5\sqrt{y}}{x} - \frac{9}{r^2}\Big) \sum_{\ell} \frac{1}{2\ell},$$

where the sum is over the even ℓ with $6x \le \ell \le L$. This sum is at least $\frac{1}{4}\ln\big(L/(6x+2)\big) \ge \frac{1}{4}\ln\big(L/(6x)\big) - 1/(12x)$, and it is at most $\ln y$. The bound for μ follows. ∎

3.2 Proof of Theorem 3.1

Let Γ and P be as in the theorem, let ξ be uniform on $\mathcal{Z}(\Gamma)$, and let r be large. Put

$$x = \log r, \qquad y = \log x, \qquad z = \log y, \qquad D = \lfloor x/10 \rfloor, \qquad L = \lfloor x\sqrt{y} \rfloor, \qquad \varepsilon = 20/z.$$

Let $\mathcal{F}$ be the event that $\xi \cap E(\Gamma - P)$ is a forest. We assume that $\mathbb{P}(\mathcal{F}) > \frac{1}{2} + \varepsilon$ and derive a contradiction.

Let $\mathcal{Q}$ be a maximal family of pairwise disjoint cycles of $\Gamma - P$ of length at most D, and let S be the union of their vertex sets. For $Q \in \mathcal{Q}$ the set $V(Q)$ is a cyclic region of Γ with $|\delta V(Q)| \le |Q| \le D$, because its vertices have degree 3. On $\mathcal{F}$ the set $\xi \cap E(\Gamma[V(Q)])$ is a forest for every Q. So Lemma 2.4 gives $|\mathcal{Q}| < (D+1)2^{D-1}/\varepsilon \le \sqrt{r}$. Hence $|S| \le D\sqrt{r} \le r/x^2$, and Proposition 3.5 applies. As $L/(6x) \ge \sqrt{y}/7$ and $W_2 \le \mu^2$, it gives

$$\mu \;\ge\; \frac{1}{4}\ln\frac{\sqrt{y}}{7} - \frac{1}{\sqrt{x}} \;\ge\; \frac{\ln y}{10}, \qquad \mathbb{E}Z^2 \;\le\; 2\mu^2 + 2\mu.$$

On $\mathcal{F}$ no cycle of $J \subseteq \Gamma - P$ is contained in ξ, so $Z = 0$. By Lemma 2.1,

$$\mathbb{P}(\mathcal{F}) \;\le\; \mathbb{P}(Z=0) \;\le\; \frac{1}{2} + \frac{1}{2\mu} \;\le\; \frac{1}{2} + \frac{5}{\ln y} \;<\; \frac{1}{2} + \frac{20}{z},$$

since $\ln y = z \ln 2$. This is a contradiction, and Theorem 3.1 is proved.

3.3 The leading term

Lemma 3.6 (Cubic multigraphs). For every $r \ge 2$ there is a connected cubic multigraph of rank r with exactly $\psi(r)$ cycles.

Proof. The numbers of cycles of the graphs of rank r are integers below 2^r, so some graph G_0 attains $\psi(r)$. If G_0 is not connected, identify a vertex of one component with a vertex of another. This changes neither the rank nor the set of cycles. Now delete a vertex of degree one, or replace a vertex of degree two and its two edges by one edge, as long as this is possible. These steps keep the rank and the number of cycles. The result is a connected multigraph with all degrees at least three, because $r \ge 2$.

Let v be a vertex of degree $d \ge 4$. Replace v by a path $v^1 v^2 \cdots v^{d-2}$, and distribute the d edge ends at v along this path: two at v^1, one at each of $v^2, \dots, v^{d-3}$, and two at v^{d-2}. The new multigraph has $d-3$ more vertices and $d-3$ more edges, so its rank is r. A cycle through v uses two edge ends at v. Together with the part of the path between these two ends it is a cycle of the new multigraph. Different cycles give different cycles, and the cycles that avoid v remain. So the number of cycles does not decrease, and it cannot exceed $\psi(r)$. Repeat this for every vertex of degree at least four. ∎

Corollary 3.7 (Leading term). There is r_1 such that every graph of rank $r \ge r_1$ has at most $\bigl(\frac{1}{2} + 21/\log^{(3)} r\bigr)2^r$ cycles. Consequently $\psi(r) \sim 2^{r-1}$.

Proof. Let r be large, and let G be a connected cubic multigraph of rank r with $\psi(r)$ cycles (Lemma 3.6). Let W be a shortest cycle of G, of length g. A breadth-first search from a vertex meets no vertex twice up to distance $\lfloor (g-1)/2 \rfloor$, so G has at least $2^{(g-2)/2}$ vertices. As G has $2r-2$ vertices, $g \le 2\log(2r) + 2$. So the set $P = V(W)$ has at most $2r/\log^2 r$ vertices, and $G[P]$ is connected. By Theorem 3.1, $q(G,P) \le \frac{1}{2} + \delta_r$ with $\delta_r = 20/\log^{(3)} r$.

Let e be an edge of W, and let C be a cycle through e. The outside part of C is a proper subset of a cycle, so it is a forest. The even subgraph $C + W$ has the same outside part, and it does not contain e. So the cycles through e give $2c_e(G)$ different even subgraphs whose outside part is a forest, and $c_e(G) \le q(G,P)\,2^{r-1}$. The other cycles of G are cycles of $G - e$, which is connected and has rank $r-1$. Hence

$$\psi(r) = c(G) \;\le\; c(G-e) + \bigl(\tfrac{1}{2} + \delta_r\bigr)2^{r-1} \;\le\; \psi(r-1) + \bigl(\tfrac{1}{2} + \delta_r\bigr)2^{r-1}.$$

Put $y_r = \psi(r)2^{-r} - \frac{1}{2}$. Then $y_r \le \frac{1}{2}y_{r-1} + \frac{1}{2}\delta_r$ for all large r. The numbers δ_r decrease, and $y_r \le \frac{1}{2}$. So $y_r \le 2^{-j}y_{r-j} + \delta_{r-j}$ for $j \le r/2$. With $j = \lfloor r/2 \rfloor$ this is at most $21/\log^{(3)} r$ when r

is large. Finally $\psi(r) > 2^{r-1}$ for $r \geq 3$ [4]; the rings of Section 6.1 also show this. ■

4 Layers

The later sections use Theorem 3.1 through Corollary 3.7 and through one lemma.

IDEA | the content of the lemma

If the outside part is a forest for more than half of the even subgraphs, by a fixed margin, then every distance layer around the marked set is small. In the other case a window of consecutive layers is a large set with a small boundary, and the forest estimate applies inside this set. We use the lemma for paths in Section 5 and for pieces in Section 7.

Let G be a multigraph and P a set of vertices. A *level function* is a map $\lambda : V(G) \to \mathbb{Z}$ such that $|\lambda(u) - \lambda(w)| \leq 1$ for every edge uw that has an end outside P. The distance from a fixed vertex is a level function. For $\delta \in (0, \frac{1}{2})$ let $R_\delta \geq 2^{100}$ be an integer such that every $r \geq R_\delta$ is in the range of Theorem 3.1 and satisfies $20/\log^{(3)} r \leq \delta$. For an integer $k \geq 1$ put

$$\Theta_\delta(k) = \max\bigl\{2R_\delta,\ 2^{100}(k+1)^2\bigr\}.$$

Lemma 4.1 (Layers). Let G be a connected subcubic multigraph, and let P be a nonempty set of vertices such that $G[P]$ is connected and every vertex outside P has degree three. Let λ be a level function such that at most k vertices of P have any given level. Let $\delta \in (0, \frac{1}{2})$, and suppose that $q(G, P) > \frac{1}{2} + \delta$. Then, for every j, fewer than $\Theta_\delta(k)$ vertices outside P have level j.

Proof. Put $\Phi(m) = m/\log^2 m$ and $\Theta = \Theta_\delta(k)$.

Step 1: no large set outside P has a small boundary. We claim that no set $X \subseteq V(G) \setminus P$ has $|X| = m \geq \Theta$ and $|\delta X| \leq \Phi(m)/4$. Suppose that X is such a set. Let C_0 be the component of $G - X$ that contains P, and put $X^+ = V(G) \setminus V(C_0)$. Then $X \subseteq X^+$, the set X^+ avoids P, and $G - X^+ = C_0$ is connected. Different components of $G - X$ have no edge between them, so $\delta X^+ \subseteq \delta X$. Every vertex of X^+ has degree three. Put $d^+ = |\delta X^+| \geq 1$.

We form a multigraph Γ from $G[X^+]$. For each edge of δX^+ add a new vertex and join it to the end of that edge in X^+. Join the d^+ new vertices by a cycle W through all of them, and let P^+ be the set of the new vertices. Every component of $G[X^+]$ has an edge to C_0, so Γ is connected. It is subcubic, $\Gamma[P^+]$ is connected, the vertices outside P^+ have degree three, and $\Gamma - P^+ = G[X^+]$. The multigraph Γ has d^+ more vertices and $2d^+$ more edges than $G[X^+]$, and $3|X^+| = 2|E(G[X^+])| + d^+$. So its rank is

$$\rho = \frac{|X^+| + d^+}{2} + 1 \ \geq\ \frac{m}{2} \ \geq\ R_\delta.$$

The function Φ is increasing on $[8, \infty)$, and $\Phi(m)/4 \leq \Phi(m/2)$. So $d^+ \leq \Phi(m)/4 \leq \Phi(m/2) \leq \Phi(\rho)$, and Theorem 3.1 with the choice of R_δ gives $q(\Gamma, P^+) \leq \frac{1}{2} + \delta$.

The set $\xi \cap E(G[X^+])$ has the same distribution for ξ uniform on $\mathcal{Z}(G)$ and for ξ uniform on $\mathcal{Z}(\Gamma)$. Indeed, contract the connected set $V(C_0)$ of G to one vertex, and delete the loops that arise. Call the result G'. An even subgraph of G restricts to an even subgraph of G'. Every even subgraph of G' arises in this way: the missing edges inside the contracted set can be chosen in a spanning tree of that set. So the restriction is a surjective linear map $\mathcal{Z}(G) \to \mathcal{Z}(G')$, and it sends the uniform distribution to the uniform distribution. The contraction of W in Γ gives the same multigraph G', with the new edges in the place of the edges of δX^+. Hence

$$\mathbb{P}\big(\xi \cap E(G[X^+]) \text{ is a forest}\big) \;=\; q(\Gamma, P^+) \;\le\; \frac{1}{2} + \delta.$$

But $G[X^+]$ is a subgraph of $G - P$, so this probability is at least $q(G,P) > \frac{1}{2} + \delta$. This contradiction proves the claim.

Step 2: a window of levels. Let m_j be the number of vertices outside P of level j. For $i \le i'$ let $X(i,i')$ be the set of vertices outside P with a level in $[i,i']$. An edge that leaves $X(i,i')$ ends outside P at level $i-1$ or $i'+1$, or it ends in P at a level in $[i-1, i'+1]$. So

$$|\delta X(i,i')| \;\le\; 3m_{i-1} + 3m_{i'+1} + 3k(i'-i+3). \tag{4.1}$$

Suppose that $m_j \ge \Theta$ for some j. Start with the window $i = i' = j$. As long as a level next to the window has more than $m/(48\log^2 m)$ vertices outside P, where $m = |X(i,i')|$ is the present size, add that level to the window. The graph is finite, so this stops. Let μ be the final size and $\gamma = \log \mu \ge 100$. Each step multiplies the size by more than $1 + 1/(48\gamma^2)$. So the number of steps is at most

$$\frac{\ln \mu}{\ln\big(1 + 1/(48\gamma^2)\big)} \;\le\; 96\gamma^2 \ln \mu \;<\; 67\gamma^3.$$

At the end both levels next to the window have at most $\mu/(48\gamma^2)$ vertices outside P. By (4.1),

$$|\delta X(i,i')| \;\le\; \frac{\mu}{8\gamma^2} + 3k\big(67\gamma^3 + 3\big) \;\le\; \frac{\mu}{4\gamma^2} = \frac{\Phi(\mu)}{4}.$$

The second inequality is equivalent to $24k\gamma^2(67\gamma^3+3) \le 2^\gamma$. This holds because $k \le 2^{\gamma/2-50}$, by $\mu \ge \Theta \ge 2^{100}(k+1)^2$, and because $24\gamma^2(67\gamma^3+3) \le 2000\gamma^5 \le 2^{\gamma/2+50}$ for $\gamma \ge 100$. So the final window contradicts Step 1. ∎

5 Poles with many paths

A cut of a cubic graph divides it into two parts, and a cycle that crosses the cut is made of paths in the two parts. So we need to count paths between prescribed ends.

IDEA | the results of this section

First, if more than half of the possible paths between two ends are present, by a fixed margin, then the part looks like a ladder, and it has few cycles (Theorem 5.4). Second, the number of paths between two ends is the largest possible, or it is smaller by a fixed factor (Theorem 5.8).

5.1 Poles

A *pole* is a connected multigraph without loops in which every vertex has degree two or three. Its vertices of degree two are its *ports*. A *k-pole* is a pole with exactly k ports. By a count of edge ends, a k-pole of rank h has $2h + k - 2$ vertices.

For two vertices $s \neq t$ of a multigraph N let $P_{st}(N)$ be the number of paths from s to t. An $\{s,t\}$-*join* is a set of edges in which s and t have odd degree and all other vertices have even degree. If N is connected of rank b, there are exactly 2^b of them: they exist, and the sum of two of them is even. Every path from s to t is an $\{s,t\}$-join, so

$$P_{st}(N) \leq 2^b. \tag{5.1}$$

Call the pair s,t *full* in N if $P_{st}(N) = 2^{\beta(N)}$.

Lemma 5.1 (Joins). Let N be a pole of rank b, and let s,t be two of its ports. Every $\{s,t\}$-join is the union of a path from s to t and of cycles, and these are pairwise vertex-disjoint. So the pair is full if and only if every cycle of N meets every path from s to t. If some cycle of N meets no path from s to t, then $P_{st}(N) \leq 2^{b-1}$. If a cycle Z of N is disjoint from some path from s to t, and $N - E(Z)$ has k components, then $P_{st}(N) \leq 2^b - 2^{b-|Z|+k-1}$.

Proof. In a join, s and t have odd degree at most two, and every other vertex has even degree at most three. So the degrees are one at s and t and zero or two elsewhere. If a cycle Z meets no path from s to t, then $J \mapsto J + Z$ maps the paths injectively to joins that are not paths. Let a cycle Z be disjoint from a path Q from s to t. Then Q is an $\{s,t\}$-join of $N - E(Z)$. So $N - E(Z)$ has $2^{\beta(N-E(Z))}$ such joins, and $\beta(N - E(Z)) = b - |Z| + k - 1$. The union of each of them with $E(Z)$ is an $\{s,t\}$-join of N that is not a path. ■

Lemma 5.2 (Reduction). Let N be a pole of rank b, and let s, t be two of its ports with $P_{st}(N) > 2^{b-1}$.

(a) Replace each port other than s, t, with its two edges, by one edge. No loop arises. The result N° is a 2-pole with ports s, t. It has the same rank, the same number of paths from s to t, and the same number of cycles as N.

(b) Every bridge of N° separates s from t and is incident with neither. Let $N_1, \dots, N_q$ be the components of N° without its bridges, in the order from s to t. Each N_i is a bridgeless 2-pole of rank $b_i \geq 1$, and

$$b = \sum_i b_i, \qquad P_{st}(N) = \prod_i P(N_i), \qquad c(N) = \sum_i c(N_i),$$

where $P(N_i)$ is the number of paths between the two ports of N_i. In particular $P(N_i)\, 2^{-b_i} \geq P_{st}(N)\, 2^{-b}$ for every i.

Proof. (a) Let v be a port other than s, t. Suppose that both edges at v end in the same vertex u. Then u has degree three, its third edge is a bridge, and no path from s to t meets the cycle through u and v. This contradicts Lemma 5.1. So the two neighbours of v are distinct, and we may replace v by an edge. Paths and cycles correspond. Repeat.

(b) Let e be a bridge of N°, and let Y be a component of $N^\circ - e$, with n vertices, j of them ports. Counting edge ends gives $\beta(Y) = (n - j + 1)/2$. This is a positive integer, so Y contains a cycle. If Y contains neither port or both ports, a cycle on one side of e meets no path from s to t. If e is incident with s, every path from s to t starts with e and avoids the cycle on the side of s. Both cases contradict Lemma 5.1. So every path from s to t passes all bridges, in the same order.

The multigraph N_i is connected and bridgeless. Its vertices have degree three, except the ends of the two bridges at N_i, or s, or t. These two vertices are distinct: a common end of two bridges would have a third edge that is a bridge and does not separate s from t. So N_i is a 2-pole, and it contains a cycle. Ranks add over the N_i, every cycle lies in one N_i, and a path from s to t is a concatenation of paths between the ports of the N_i and of the bridges. The last statement holds because $P(N_j) \leq 2^{b_j}$ for all j. ■

5.2 The shape of a pole with many paths

Let N be a bridgeless 2-pole with ports s, t and rank $b \geq 2$. It has $2b$ vertices. It has no cut vertex: a cut vertex of a subcubic multigraph is joined to some component by a single edge. We use the following notation. The *layer* L_j is the set of vertices at distance j from s. Every edge joins vertices of the same layer or of consecutive layers. Let ℓ be the distance from s to t, let $A = w_0 w_1 \cdots w_\ell$ be a shortest path from s to t, and put $J = N - V(A)$. Then $w_j \in L_j$. A vertex of A lies on two edges of A, or on one if it is a port. So it has at most one *further edge*. This edge leads to J, or it is parallel to an edge of A, because A is a shortest path. For $0 < j < \ell$ the layer L_j separates s from t, so $|L_j| \geq 2$. Call L_j *thin* if $0 < j < \ell$ and $|L_j| = 2$. A thin layer consists of w_j and one vertex of J. Call a thin layer *ordinary* if its vertex of J is on no pair of parallel edges

(Figure 5).

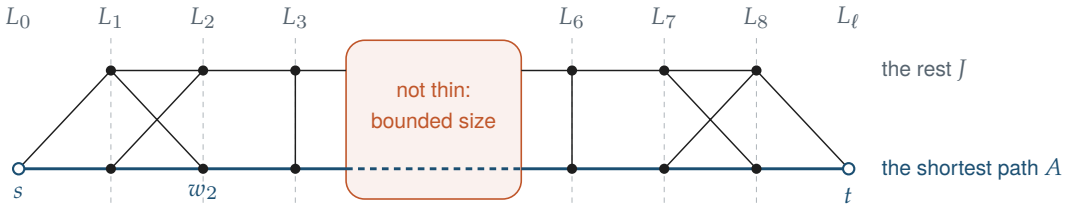


Figure 5. A bridgeless 2-pole with many paths from s to t (schematic). Almost every layer is thin: it has one vertex of the shortest path A and one vertex of J (Proposition 5.3).

IDEA | outline of the proof of Proposition 5.3

Add the shortest path A to a path from s to t. The result is an even subgraph whose edges away from A form a forest. So the outside part is a forest for more than half of the even subgraphs, and every layer is small by Lemma 4.1. By Lemma 2.4 there are few short cycles away from A. By a count of vertices, almost every layer then has exactly two vertices.

Proposition 5.3 (Thin layers). For every $\varepsilon \in (0, \frac{1}{2}]$ there is a number $C = C_\varepsilon \geq 1$ with the following property. Let N be a bridgeless 2-pole of rank $b \geq 2$ with ports s, t and $P_{st}(N) \geq (\frac{1}{2} + \varepsilon)2^b$. Then, in the notation above,

(i) $\beta(J) \leq C$ and $b - C \leq \ell \leq b$;

(ii) at most $2C$ of the layers L_j with $0 < j < \ell$ are not thin, and $L_j = \emptyset$ for $j > b + C$;

(iii) at most $10C + 2$ vertices of N lie outside the ordinary layers.

Proof. Step 1: the layers are small. Put $P = V(A)$. Then $N[P]$ is connected, and every vertex outside P has degree three. For a path Q from s to t, the set $Q + A$ is an even subgraph of N. Its edges in $N - P$ are edges of the path Q, so they form a forest. Different paths give different even subgraphs, and N has 2^b even subgraphs. Hence

$$q(N, P) \;\geq\; \frac{P_{st}(N)}{2^b} \;\geq\; \frac{1}{2} + \varepsilon. \tag{5.2}$$

Let $\lambda(u)$ be the distance from s to u in N. This is a level function, and P has at most one vertex of each level. By Lemma 4.1 with $\delta = \varepsilon/2$, every layer L_j has fewer than $\Theta := \Theta_{\varepsilon/2}(1)$ vertices of J.

Step 2: a short cycle. Put $U_j = L_j \setminus V(A)$, and for an interval I of indices let J_I be the subgraph of J induced by the sets U_j with $j \in I$. Suppose that J_I contains a cycle. Choose a cycle Z of J_I such that the interval $[a, a']$ of the layers that Z meets is as short as possible, and put $\sigma = a' - a$ and $R = U_a \cup \cdots \cup U_{a'}$. We claim that $|R| \leq 20\Theta$.

By the choice of Z, every cycle of $J[R]$ meets U_a. So $J[R] - U_a$ is a forest. Deleting a vertex of

degree at most three lowers the rank by at most two, so $\beta(J[R]) \le 2|U_a| < 2\Theta$. The vertices of R have degree three in N. Counting edge ends gives $|R| = 2\beta(J[R]) + |\delta R| - 2\,\mathrm{comp}(J[R])$. An edge that leaves R ends in U_{a-1}, in $U_{a'+1}$, or in one of the vertices $w_{a-1}, \dots, w_{a'+1}$, each of which has at most one further edge. So $|\delta R| < 6\Theta + \sigma + 3$, and $|R| < 10\Theta + \sigma + 1$. The cycle Z has a vertex in layer a and a vertex in layer a', so $|Z| \ge 2\sigma$. Also $|Z| \le |R|$. Hence $\sigma \le 10\Theta$ and $|R| \le 20\Theta$.

Step 3: the rank of J. Start with the interval of all layers. As long as J_I contains a cycle for some interval I of the present family, choose Z and R as in Step 2, and replace I by the at most two intervals that remain when $[a, a']$ is removed. Let $Z_1, \dots, Z_k$ be the chosen cycles and $R_1, \dots, R_k$ the corresponding sets. The sets R_i are pairwise disjoint, $|R_i| \le 20\Theta$, and $J - \bigcup_i R_i$ is a forest, because an edge of J joins equal or consecutive layers. So $\beta(J) \le 2\sum_i |R_i| \le 40\Theta k$.

The sets $V(Z_i)$ are pairwise disjoint cyclic regions of N that avoid P. Each has at most $|Z_i| \le 20\Theta$ edges that leave it. If $\xi \cap E(N - P)$ is a forest, then $\xi \cap E(N[V(Z_i)])$ is a forest for every i. So Lemma 2.4 and (5.2) give $k \le (20\Theta + 1)2^{20\Theta - 1}/\varepsilon$. Hence $\beta(J) \le C$ with

$$C = C_\varepsilon := \frac{40\Theta\,(20\Theta + 1)\,2^{20\Theta - 1}}{\varepsilon}.$$

Step 4: counting. The layers $L_0, \dots, L_\ell$ contain at least 2ℓ vertices, so $\ell \le b$. The vertices of A meet the ℓ edges of A and at most $\ell + 1$ other edges. If J is not empty, then $|E(J)| \ge (3b - 1) - (2\ell + 1)$ and $|V(J)| = 2b - \ell - 1$, so $\beta(J) \ge |E(J)| - |V(J)| + 1 \ge b - \ell$. If J is empty, then $2b = \ell + 1$. In both cases $\Delta := b - \ell \le C$. This is (i).

Compare N with a multigraph that has one vertex in L_0 and in L_ℓ, two in each layer between, and none beyond. Then N has exactly 2Δ vertices more. So at most 2Δ layers between L_0 and L_ℓ are not thin, and at most 2Δ layers lie beyond L_ℓ. This is (ii). A layer between L_0 and L_ℓ with $w \ge 3$ vertices has $w - 2 \ge w/3$ vertices too many. So at most $2 + 3 \cdot 2\Delta$ vertices lie outside the thin layers.

Finally let $y \in J$ be on a pair of parallel edges. The other end of the pair has two edges to y, so it is not on A, whose vertices have at most one further edge. So the pair is a cycle of J. Two such pairs have no common vertex, because N is subcubic and has more than two vertices. So there are at most $\beta(J) \le C$ pairs, and at most $2C$ thin layers are not ordinary. They contain at most $4C$ vertices. This gives (iii). ∎

Theorem 5.4 (Few cycles). For every $\varepsilon \in (0, \frac{1}{2}]$ there is a constant K_ε with the following property. Let N be a pole of rank b, and let s, t be two of its ports with $P_{st}(N) \ge (\frac{1}{2} + \varepsilon)2^b$. Then

$$c(N) \;\le\; K_\varepsilon\, 2^{b/2}.$$

Proof. Let $C = C_\varepsilon$ be as in Proposition 5.3. We prove that a bridgeless 2-pole with these properties has at most $2^{31C+22}\,2^{b/2}$ cycles. The theorem follows with $K_\varepsilon = 2^{31C+23}$. Indeed,

in the notation of Lemma 5.2, each N_i has the required number of paths, and $\sum_i 2^{b_i/2} \le 2^{b/2} + q - 1 \le 2 \cdot 2^{b/2}$, because $x + y \le xy + 1$ for $x, y \ge 1$.

So let N be a bridgeless 2-pole. If $b = 1$, then N has one cycle. Let $b \ge 2$, and use the notation above. Let Z be a cycle, and let L_i and L_j be the first and the last layer that it meets. Let M be the set of the vertices that lie in L_i, in L_j, or outside the ordinary layers. By Proposition 5.3(iii), $|M| \le 10C + 6$.

Two strands. Let $L_l = \{w_l, y_l\}$ be an ordinary layer with $i < l < j$. The cycle Z has vertices before and after this layer. So it has at least two edges from L_{l-1} to L_l and at least two edges from L_l to L_{l+1}. The two vertices of L_l have degree at most two in Z. So Z has exactly two edges on each side of L_l and does not contain an edge $w_l y_l$. Suppose that both edges from L_{l-1} end in the same vertex of L_l. Then both edges to L_{l+1} start in the other vertex, and no edge of Z joins the part of Z before L_l to the part after it. This is impossible, because Z is connected. So each vertex of L_l has one edge of Z to L_{l-1} and one to L_{l+1}.

Matchings. Let L_l and L_{l+1} be ordinary layers with $i < l$ and $l + 1 < j$. By the last paragraph, the edges of Z between them form a perfect matching between $\{w_l, y_l\}$ and $\{w_{l+1}, y_{l+1}\}$. We count the perfect matchings. The edge $y_l y_{l+1}$ is not doubled, because the layers are ordinary. The edges $w_l y_{l+1}$ and $y_l w_{l+1}$ are not doubled, because a vertex of A has at most one further edge. If the edge $w_l w_{l+1}$ of A is doubled, then w_l and w_{l+1} have no edge to J. So there are at most two perfect matchings. If there are two, then w_{l+1} has two edge ends towards L_l: either both w_l and y_l are its neighbours, or $w_l w_{l+1}$ is doubled. In the same way w_{l+1} has two edge ends towards L_{l+2} if there are two perfect matchings between L_{l+1} and L_{l+2}. Both cannot happen, because w_{l+1} has degree three. So among consecutive pairs of ordinary layers, no two neighbouring pairs have two perfect matchings each.

The count. An edge of Z has an end in M, or it joins two consecutive ordinary layers strictly between L_i and L_j. So Z is determined by i and j, by its edges at M, and by its perfect matchings. There are at most $2^{3|M|}$ sets of edges at M. By the last paragraph there are at most $2^{(j-i)/2}$ choices for the matchings. All layers have an index of at most $n := b + C$, by Proposition 5.3(ii). Hence

$$c(N) \le 2^{30C+18} \sum_{0 \le i \le j \le n} 2^{(j-i)/2} \le 2^{30C+18} \cdot 12 \cdot 2^{(b+C)/2} \le 2^{31C+22}\, 2^{b/2}. \qquad \blacksquare$$

Consequences of Theorem 5.4. From now on fix

$$\varepsilon_0 = \tfrac{1}{1000}, \qquad K_0 = K_{\varepsilon_0}, \qquad \kappa(h) = 2K_0\, 2^{-h/2}.$$

For four distinct ports a, b, c, d of a pole N let $f_{ab|cd}(N)$ be the number of pairs of vertex-disjoint paths of N, one from a to b and one from c to d.

Lemma 5.5. Let N be a pole of rank h. In each statement below, the letters denote distinct ports of N. Write P_{xy} for $P_{xy}(N)$ and f for $f(N)$.

(a) If $P_{ab} \geq 0.501 \cdot 2^h$, then $c(N) \leq K_0\, 2^{h/2}$.

(b) If $P_{ac} + P_{bc} \geq 1.002 \cdot 2^h$, then $P_{ab} \leq \kappa(h)\, 2^h$.

(c) If $P_{ac} + P_{ad} + P_{bc} + P_{bd} \geq 2.004 \cdot 2^h$, then $f_{ac|bd} + f_{ad|bc} \leq \kappa(h)\, 2^h$.

(d) If $P_{ab} + f_{ac|bd} + f_{ad|bc} \geq 1.002 \cdot 2^h$, then $P_{cd} \leq \kappa(h)\, 2^h$.

Proof. (a) is Theorem 5.4. In each of the other cases we change N into a pole N' and apply Theorem 5.4 to two ports of N'.

(b) Add a new vertex z and join it to a and b. Then N' has rank $h+1$, and $P_{zc}(N') = P_{ac} + P_{bc} \geq (\frac{1}{2} + \varepsilon_0)2^{h+1}$. The cycles of N' through z correspond to the paths of N from a to b. So $P_{ab} \leq c(N') \leq K_0\, 2^{(h+1)/2}$.

(c) Add a new vertex z joined to a and b, and a new vertex w joined to c and d. Then N' has rank $h+2$, and $P_{zw}(N')$ is the sum of the four path counts. The cycles of N' through z and w correspond to the pairs counted by $f_{ac|bd} + f_{ad|bc}$. So this sum is at most $K_0\, 2^{(h+2)/2}$.

(d) Add an edge between c and d. Then N' has rank $h+1$. A path from a to b in N' avoids the new edge, or it consists of the new edge and a pair counted by $f_{ac|bd} + f_{ad|bc}$. So $P_{ab}(N') = P_{ab} + f_{ac|bd} + f_{ad|bc}$. The cycles of N' through the new edge correspond to the paths of N from c to d. So $P_{cd} \leq K_0\, 2^{(h+1)/2}$. ∎

5.3 Full pairs

A *square* is a copy of the complete bipartite graph $K_{2,2}$; its two classes are its *left pair* and its *right pair*. A *single rung* is one edge; its two ends form both its left pair and its right pair. These are the two kinds of *tiles*. The *weight* of a square is 2, and the weight of a single rung is 1. A *chain* of tiles $\tau_1, \ldots, \tau_m$ consists of these tiles, pairwise disjoint, and of two disjoint edges from the right pair of τ_i to the left pair of τ_{i+1} for $1 \leq i < m$. These two edges are an *interface*. A *capped ladder* is a chain with two more vertices s and t, the *caps*: s is joined to the two vertices of the left pair of τ_1, and t to the two vertices of the right pair of τ_m (Figure 6). A capped ladder is a 2-pole. If its tiles have total weight W, it has $2W+2$ vertices and rank $W+1$.

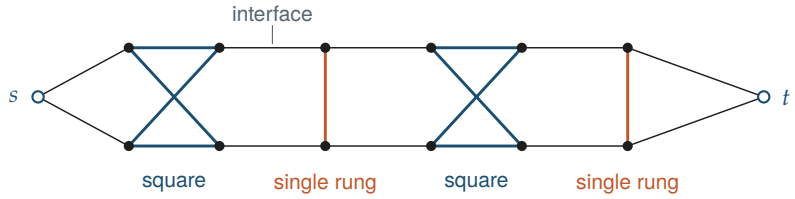


Figure 6. A capped ladder with four tiles, of weight 6 and rank 7. Without s and t it is a chain. If the right pair of the last tile is joined to the left pair of the first tile instead, the result is a ring (Section 6.1).

Lemma 5.6 (Ladders). Let N be a bridgeless 2-pole whose pair of ports is full. Then N consists of two vertices and two parallel edges, or N is a capped ladder whose caps are the ports.

Proof. Let s, t be the ports and b the rank. If $b = 1$, then N is a cycle with two vertices. Let $b \geq 2$, and use the notation of Section 5.2. By Lemma 5.1, every cycle meets the path A, so J is a forest. The count in Step 4 of the proof of Proposition 5.3 gives $\ell \leq b$. It also shows that J is not empty, because $2b > b + 1$, and that $b - \ell \leq \beta(J) = 0$. So $\ell = b$, and equality holds in every estimate of that count. This means the following. Every layer L_j with $0 < j < b$ consists of w_j and of one vertex y_j of J, and there are no other vertices outside A. Every vertex of A has a further edge, and this edge leads to J. And J is a tree.

An edge joins vertices of equal or consecutive layers. So J is the path $y_1 y_2 \cdots y_{b-1}$, the further edge of s leads to y_1, that of t leads to y_{b-1}, and the further edge of w_j leads to a vertex $y_{\pi(j)}$ with $|\pi(j) - j| \leq 1$, for $0 < j < b$. Every y_i has degree three, so it is the end of exactly one of the edges $w_j y_{\pi(j)}$. Hence π is a permutation of $\{1, \ldots, b-1\}$ that moves every number by at most one. So π is composed of fixed points and of exchanges of two consecutive numbers. A fixed point j is a single rung with the ends w_j, y_j. An exchange of j and $j + 1$ is a square with the left pair w_j, y_j and the right pair w_{j+1}, y_{j+1}. The remaining edges of A and J are the interfaces and the edges at the caps. ∎

We need one count in a capped ladder. Let $\tau_1, \ldots, \tau_m$ be its tiles, let $1 \leq i \leq m$, and let u be a vertex of the left pair of τ_i. The number of paths from u to t inside $\tau_i \cup \cdots \cup \tau_m \cup \{t\}$ is 2^q, where q is the weight of $\tau_i, \ldots, \tau_m$. Indeed, such a path passes the tiles in order. It enters a tile at a vertex of the left pair and leaves it at a vertex of the right pair. In a square there are two paths to each vertex of the right pair. In a single rung the path stays or crosses.

Lemma 5.7 (A third port). Let T be a 3-pole of rank h with ports s, t, v. If the pair s, t is full, then $P_{vt}(T) = 2^h$ or $P_{vt}(T) \leq \frac{7}{8} 2^h$.

Proof. By Lemma 5.2, T arises from a 2-pole N with ports s, t when an edge e is subdivided by v, and N consists of bridges and of bridgeless 2-poles $N_1, \ldots, N_q$ with full pairs. If e is a bridge or an edge of some N_i with $i \geq 2$, then no path from v to t meets N_1. Such a path lies in a part of T of rank at most $h - 1$, so $P_{vt}(T) \leq 2^{h-1}$ by (5.1). Let e be an edge of N_1, and let t_1 be the second port of N_1. A path from v to t is a path from v to t_1 followed by paths through $N_2, \ldots, N_q$, and each of these 2-poles is full. So we may assume that $q = 1$.

By Lemma 5.6, N is a cycle with two vertices or a capped ladder. In the first case T is a triangle and $P_{vt} = 2 = 2^h$. In the second case let $\tau_1, \ldots, \tau_m$ be the tiles, of total weight $W = h - 1$, and let a, b be the left pair of τ_1. There are four cases.

The edge e is at s. Let $e = sa$. A path from v to t starts with va or with vsb, and then it continues inside the tiles. So $P_{vt} = 2 \cdot 2^W = 2^h$.

The edge e is τ_1, a single rung. A path from v goes to a or to b. From a it continues to the right, or it passes s and b and continues to the right from b. In both cases there are 2^{W-1} continuations. The same holds for b. So $P_{vt} = 4 \cdot 2^{W-1} = 2^h$.

The edge e is in τ_1, a square. Let a', b' be the right pair, and let $e = aa'$. The cycle $Z = sab'b$ is disjoint from a path that starts with va' and goes on to the right. In $T - E(Z)$ the vertex s is isolated, and all other vertices are in one component. By Lemma 5.1, $P_{vt} \le 2^h - 2^{h-3} = \frac{7}{8}\, 2^h$.

The edge e is not at s and not in τ_1. Let Z be the square τ_1, or the triangle sab if τ_1 is a single rung. Some path from v to t goes from v to the right and does not meet $\{s\} \cup \tau_1$. The multigraph $T - E(Z)$ has two components. One of them consists of s, and also of a and b if τ_1 is a square. By Lemma 5.1, $P_{vt} \le 2^h - 2^{h-|Z|+1} \le \frac{7}{8}\, 2^h$. ∎

Theorem 5.8 (Gap). Let N be a pole of rank b, and let s, t be two of its ports. Then

$$P_{st}(N) = 2^b \qquad \text{or} \qquad P_{st}(N) \le \tfrac{15}{16}\, 2^b.$$

Proof. We use induction on b. Let $P_{st}(N) > 2^{b-1}$. By Lemma 5.2 we may assume that N is a 2-pole. If N has a bridge, the claim holds for the multigraphs N_i of Lemma 5.2, which have smaller rank, and the path counts multiply. So let N be bridgeless. If $b = 1$, the pair is full. Let $b \ge 2$. If s and t are adjacent, then $P_{st}(N) \le 1 + 2^{b-1} \le \frac{3}{4}\, 2^b$, by (5.1) for N without that edge. Otherwise let x and y be the neighbours of s. They are distinct, because N has no bridge. The multigraph $T = N - s$ is connected, because N has no cut vertex. It is a 3-pole of rank $b - 1$ with ports x, y, t, and

$$P_{st}(N) = P_{xt}(T) + P_{yt}(T).$$

By induction each term is 2^{b-1} or at most $\frac{15}{16} 2^{b-1}$. If neither pair is full, we are done. If the pair x, t is full, then $P_{yt}(T)$ is 2^{b-1} or at most $\frac{7}{8}\, 2^{b-1}$, by Lemma 5.7. So $P_{st}(N)$ is 2^b or at most $\frac{15}{16} 2^b$. ∎

Corollary 5.9 (Four ports). Let N be a 4-pole without bridge, with ports a, b, c, d such that the four pairs ac, ad, bc, bd are full. Then N is a chain of tiles; a, b is the left pair of its first tile and c, d is the right pair of its last tile.

Proof. Add a vertex z joined to a, b and a vertex w joined to c, d. The result N' is a 2-pole of rank $\beta(N) + 2$ with ports z, w, and $P_{zw}(N') = 4 \cdot 2^{\beta(N)}$. So the pair z, w is full. No edge of N is a bridge of N', because N has no bridge. No edge at z is a bridge, because the second edge at z ends in another vertex of N; the same holds for w. So N' has no bridge, and it has more than two vertices. By Lemma 5.6, N' is a capped ladder with the caps z and w. The neighbours a, b of z form the left pair of its first tile, and the neighbours c, d of w form the right pair of its last tile. Without the caps, this is a chain. ∎

6 Rings and cyclic cuts of size at most four

In this section G is a connected cubic multigraph without loops, of rank r.

IDEA | the results of this section

We count the cycles of the rings. Then we show that a graph of large rank with a cyclic cut of size at most four is a ring, or it has at most $2^{r-1} + O(2^{r/2})$ cycles (Proposition 6.3 and Theorem 6.8). The proof uses the gap of Section 5: a side of the cut has all possible paths between two ends, or a fixed proportion of them is missing. For a cut of size four it also uses a *cap*: a small tree that closes the larger side of the cut (Lemma 6.5).

Let X be a set of $n \geq 1$ vertices of G with $|\delta X| = d$. Counting edge ends gives $\beta(G[X]) = (n-d)/2 + \mathrm{comp}(G[X])$. So $G[X]$ contains a cycle if $d \leq 2$, and also if $d = 3$ and $n \geq 2$. Hence every cut of size at most two is cyclic, and a cut of size three is cyclic unless one of its sides is a single vertex.

6.1 Rings

A *ring* consists of pairwise disjoint tiles $\tau_1, \ldots, \tau_m$ and of two disjoint edges from the right pair of τ_i to the left pair of τ_{i+1} for every i, with indices modulo m. These pairs of edges are the *interfaces*. Let the ring have a squares and s single rungs, and let $W = 2a + s$ be its weight. We assume that $W \geq 7$. Then the ring is a cubic graph with $2W$ vertices and rank $W + 1$. Put

$$E(a,s) = 2^a\left(2a + 4s + \tbinom{s}{2} + \tfrac{1}{2}\right) - 3a - 4s + \tbinom{s}{2}.$$

Then $2^{2q} + E(q,0)$ and $2^{2q+1} + E(q,1)$ are the numbers $H(2q+1)$ and $H(2q+2)$ of (1.1).

Proposition 6.1 (Rings). A ring with $a \geq 1$ squares and s single rungs, of weight $W \geq 7$, has at most $2^W + E(a,s)$ cycles. Equality holds if $s \leq 1$. A ring of weight $W \geq 7$ without squares has at most $2^W + W^2$ cycles.

Proof. Two interfaces together form a cut. So a cycle uses an odd number of edges of every interface, or an even number of edges of every interface.

One edge of every interface. There are 2^m ways to choose the edges. The cycle then enters each tile at one vertex of its left pair and leaves it at one vertex of its right pair. In a square there are two such paths. In a single rung there is one: the cycle stays at the vertex or crosses the rung. Every choice gives a cycle. So there are $2^{m+a} = 2^W$ cycles of this kind.

No interface. These cycles are the a squares.

Both edges of every interface. In a square the cycle then uses two disjoint edges, and there are two ways to choose them. In a single rung it does not use the rung. Each of the 2^a choices is a cycle or a union of two cycles. Changing the choice in one square changes one into the other. So there are 2^{a-1} such cycles if $a \geq 1$, and at most one if $a = 0$.

Both edges of some interfaces, but not of all. Let the cycle use the interface after τ_i and not the

interface before τ_i. Then it turns in τ_i: it joins the two vertices of the right pair inside the tile. It passes the next tiles on two disjoint paths, until it turns in a tile τ_j. This is already a closed curve, so the cycle uses exactly the interfaces from τ_i forward to τ_j. In each tile from τ_i to τ_j it has two possibilities if the tile is a square, and one if it is a single rung, in either role. Every choice gives a cycle. Let $I(i,j)$ be the set of the other tiles, those strictly after τ_j and strictly before τ_i. Then the number of these cycles is $2^a\,\omega(i,j)$, where $\omega(i,j) = 2^{-n}$ and n is the number of squares in $I(i,j)$. Every cycle of the present kind arises for exactly one ordered pair (i,j) of distinct tiles.

So a ring with $a \geq 1$ has $2^W + a + 2^{a-1} + 2^a\,\Omega$ cycles, where Ω is the sum of $\omega(i,j)$ over the ordered pairs of distinct tiles. We compute Ω. Let τ_j be a square. The squares $\tau_i \neq \tau_j$ give $\omega = 1, \frac{1}{2}, \dots, 2^{2-a}$, in the order in which they follow τ_j. So the pairs of two squares contribute $a(2 - 2^{2-a})$. Let τ_i be a single rung. The squares τ_j give $\omega = 1, \frac{1}{2}, \dots, 2^{1-a}$, in the order in which they precede τ_i. The same holds with the roles of i and j exchanged. So the pairs of a single rung and a square contribute $2s(2 - 2^{1-a})$. Finally let τ_i, τ_j be two single rungs, with n squares on one side between them and $a - n$ on the other. The two ordered pairs contribute $2^{-n} + 2^{-(a-n)} \leq 1 + 2^{-a}$, with equality if and only if $n = 0$ or $n = a$. Hence

$$\Omega \;\leq\; 2a + 4s + \tbinom{s}{2} - 2^{-a}\Big(4a + 4s - \tbinom{s}{2}\Big),$$

with equality if $s \leq 1$, because then there is no pair of two single rungs. This is the first statement.

If $a = 0$, every ordered pair of distinct tiles gives one cycle. There are $W(W-1)$ pairs, and at most one further cycle uses both edges of all interfaces. ∎

Corollary 6.2. Let $r \geq 8$. Every ring of rank r has at most $H(r)$ cycles. A ring of rank r with $\lfloor (r-1)/2 \rfloor$ squares and at most one single rung has exactly $H(r)$ cycles.

Proof. Let $W = r - 1$. A ring of weight W with $a = \lfloor W/2 \rfloor$ squares has $s = W - 2a \leq 1$ single rungs, and it has $2^W + E(a,s) = H(r)$ cycles by Proposition 6.1. For $a \geq 2$ and $2a + s = W$, a direct computation gives

$$E(a-1, s+2) - E(a,s) = -2^{a-1}\Big(W + \frac{s^2 + s}{2} - \frac{13}{2}\Big) + 2s - 4 \;<\; 0,$$

because $W \geq 7$. So $2^W + E(a,s) \leq H(r)$ for every ring of weight W with a square. The number $H(r) - 2^W$ is 71 for $W = 7$ and 124 for $W = 8$, and it more than doubles when W grows by two. So it exceeds W^2, and a ring without squares has fewer than $H(r)$ cycles. ∎

We write R_r for a ring of rank $r \geq 8$ with $\lfloor (r-1)/2 \rfloor$ squares and at most one single rung. In the notation of Horak and Kim [5] it is the graph MC_{r-1}, and they computed $c(R_r) = H(r)$.

6.2 Cuts of size at most three

Proposition 6.3. There is a constant r_3 with the following property. Let G be a connected cubic multigraph without loops, of rank $r \geq r_3$, that has a cyclic cut of size at most three. Then

$$c(G) \;\leq\; 2^{r-1} + 5K_0^2\, 2^{r/2}.$$

Proof. Let $d \leq 3$ be the least size of a cyclic cut, and let δX be a cyclic cut of size d. By the remark at the start of this section, G has no cut of size less than d. So $G[X]$ and $G - X$ are connected, and no vertex is on two edges of δX if $d \geq 2$: otherwise we could move that vertex to the other side and get a smaller cut. Let T_1 and T_2 be the two sides, with ranks $b_1 \leq b_2$. Then $b_1 \geq 1$ and $r = b_1 + b_2 + d - 1$.

If $d = 1$, then $c(G) = c(T_1) + c(T_2) \leq (2^{b_1} - 1) + (2^{b_2} - 1) \leq 2^{r-1}$, because $b_1 + b_2 = r$.

Let $d \in \{2, 3\}$. Then T_1 and T_2 are d-poles, and the cut edges match their ports. A cycle that crosses the cut uses exactly two cut edges, and it consists of these and of one path in each side. For a pair j of cut edges let $x_j\, 2^{b_1}$ and $a_j\, 2^{b_2}$ be the numbers of paths between the corresponding ports of T_1 and of T_2. Then $0 \leq x_j, a_j \leq 1$ by (5.1), and

$$c(G) \;=\; c(T_1) + c(T_2) + 2^{b_1+b_2} \sum_j x_j a_j, \qquad 2^{r-1} = 2^{d-2}\, 2^{b_1+b_2}. \tag{6.1}$$

There is one pair for $d = 2$, and there are three pairs for $d = 3$. Let r be large. Then $b_2 \geq (r-2)/2$ is large, and $\kappa(b_2) \leq \frac{1}{1000}$. We have $c(T_1) < 2^{b_1}$.

Case A: $a_j \leq 0.501$ **for all** j. By Corollary 3.7, $c(T_2) \leq \frac{9}{16} 2^{b_2} \leq \frac{9}{32} 2^{b_1+b_2}$. So $c(G)\, 2^{-b_1-b_2}$ is less than $2^{-b_2} + \frac{9}{32} + 0.501$ for $d = 2$, and less than $2^{-b_2} + \frac{9}{32} + 1.503$ for $d = 3$. In both cases $c(G) < 2^{r-1}$.

Case B: $a_j > 0.501$ **for some** j. Then $c(T_2) \leq K_0\, 2^{b_2/2}$ by Lemma 5.5(a). So $c(T_1) + c(T_2) < 2^{b_1} + K_0\, 2^{b_2/2} \leq \frac{1}{1000} 2^{b_1+b_2}$.

Let $d = 2$. If $x_1 < 0.501$, then $c(G) < 0.502 \cdot 2^{b_1+b_2} < 2^{r-1}$ by (6.1). Otherwise $c(T_1) \leq K_0\, 2^{b_1/2}$ by Lemma 5.5(a), and

$$c(G) \;\leq\; 2^{r-1} + K_0\big(2^{b_1/2} + 2^{b_2/2}\big) \;\leq\; 2^{r-1} + 2K_0\, 2^{r/2}.$$

Let $d = 3$, and number the pairs so that $a_1 \geq a_2 \geq a_3$. Any two pairs of ports of a 3-pole have one port in common. If $a_1 + a_2 < 1.002$, then $\sum_j x_j a_j < 1.503$ and $c(G) < 2^{r-1}$. Otherwise $a_3 \leq \kappa(b_2)$ by Lemma 5.5(b). If $x_1 + x_2 < 1.002$, then $\sum_j x_j a_j < 1.003$ and $c(G) < 2^{r-1}$. Otherwise $x_3 \leq \kappa(b_1)$ and $c(T_1) \leq K_0\, 2^{b_1/2}$ by Lemma 5.5(b) and (a). Then (6.1) gives

$$c(G) \;\leq\; 2K_0\, 2^{b_2/2} + 2^{b_1+b_2}\big(2 + \kappa(b_1)\kappa(b_2)\big) \;=\; 2^{r-1} + 2K_0\, 2^{b_2/2} + 4K_0^2\, 2^{(b_1+b_2)/2} \;\leq\; 2^{r-1} + 5K_0^2\, 2^{r/2}.$$

■

6.3 States of a pole with four ports

Let Y be a 4-pole of rank h with ports $1, 2, 3, 4$. A *pairing* is one of the three partitions $12|34$, $13|24$, $14|23$ of the ports into two pairs; the two pairs of a pairing are *opposite*. For a pair ij and a pairing $\alpha = ij|kl$ put

$$a_{ij} = 2^{-h} P_{ij}(Y), \qquad u_\alpha = 2^{-h} f_{ij|kl}(Y), \qquad S = \sum_{ij} a_{ij}, \qquad F = \sum_{\alpha} u_\alpha.$$

A pair of disjoint paths that joins the four ports is a join for the set of all four ports, and there are 2^h such joins. So $a_{ij} \le 1$ and $F \le 1$. We write $a_{ij}(Y)$, $S(Y)$ and so on when several poles are present. In this notation Lemma 5.5 reads as follows, for distinct ports i, j, k, l and with $\kappa = \kappa(h)$:

$$a_{ik} + a_{jk} \ge 1.002 \implies a_{ij} \le \kappa; \tag{6.2}$$

$$a_{ik} + a_{il} + a_{jk} + a_{jl} \ge 2.004 \implies u_{ik|jl} + u_{il|jk} \le \kappa; \tag{6.3}$$

$$a_{kl} > \kappa \implies a_{ij} + F - u_{ij|kl} < 1.002. \tag{6.4}$$

Fix h_0 such that $\kappa(h) \le \frac{1}{1000}$ for $h \ge h_0$.

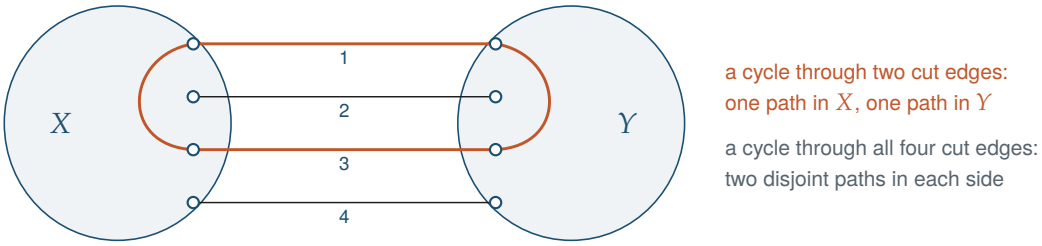


Figure 7. Two 4-poles X and Y, joined at their ports. The number a_{ij} counts the paths between the ports i and j of one side, and u_α counts the pairs of disjoint paths for a pairing α of the ports, both as a proportion of the largest possible number.

Lemma 6.4 (Gluing). Let X and Y be 4-poles of ranks h and k, and let G arise from them when we join each port of X to the port of Y with the same number. Then $\beta(G) = h + k + 3$ and

$$c(G) = c(X) + c(Y) + 2^{h+k}\,\mathcal{B}, \qquad \mathcal{B} = \sum_{ij} a_{ij}(X)\, a_{ij}(Y) + \sum_{\alpha \ne \beta} u_\alpha(X)\, u_\beta(Y),$$

and $\mathcal{B} \le S(Y) + F(Y) - \min_\alpha u_\alpha(Y)$.

Proof. A cycle that crosses the cut uses two or four of the new edges. With two, at the ports i and j, it consists of a path between these ports in each pole. With four, it consists of a pair of disjoint paths in each pole. If the pairings in X and Y are equal, the union is two cycles. If they differ, it is one cycle. For the inequality, $a_{ij}(X) \le 1$, and

$$\sum_{\alpha \ne \beta} u_\alpha(X)\, u_\beta(Y) = \sum_{\alpha} u_\alpha(X)\big(F(Y) - u_\alpha(Y)\big) \;\le\; \max_\alpha \big(F(Y) - u_\alpha(Y)\big),$$

because the numbers $u_\alpha(X)$ are not negative and their sum $F(X)$ is at most 1. ■

Lemma 6.5 (Cap). There is $h_1 \geq h_0$ with the following property. Let Y be a 4-pole of rank $h \geq h_1$, and put $z = 2^{-h}c(Y)$. Then, for every pairing α,

$$z + S + F - u_\alpha \leq 4.001.$$

Proof. Let $\alpha = ij|kl$. Add two new vertices x and y and an edge between them, join x to the ports i and j, and join y to the ports k and l. The new multigraph $\hat{Y}$ has two more vertices and five more edges than Y, so its rank is $h+3$. The new edges form a tree, and this tree joins any two ports by a path. A path of Y between two ports and the path of the tree between its ends form a cycle. A pair of disjoint paths of Y whose pairing is not α and the four edges of the tree at the ports form a cycle. With the cycles of Y, these are $c(Y) + (S + F - u_\alpha)2^h$ distinct cycles of $\hat{Y}$. By Corollary 3.7, $\hat{Y}$ has at most $(\frac{1}{2} + 2^{-13})\, 2^{h+3} = (4 + 2^{-10})\, 2^h$ cycles, when h is large. ■

Lemma 6.6 (Rigidity). Let Y be a 4-pole of rank $h \geq h_0$ with $S + F \geq 3.99$. Then three pairs with a common port are full, or the four pairs between two disjoint pairs of ports are full.

Proof. For a partition of the ports into two pairs, the *cross sum* is the sum of a_{ij} over the four pairs ij that are not pairs of the partition. Each pair ij occurs in two of the three cross sums. We use Theorem 5.8 in this form: if $a_{ij} > \frac{15}{16}$, then ij is full.

Case 1: some cross sum is at least 2.004. Let it be the cross sum X of $12|34$. Then $a_{13} + a_{23} \geq 1.002$ or $a_{14} + a_{24} \geq 1.002$, so $a_{12} \leq \kappa$ by (6.2). In the same way $a_{34} \leq \kappa$. By (6.3), $u_{13|24} + u_{14|23} \leq \kappa$. Put $u = u_{12|34}$. Then

$$S + F \leq X + u + 3\kappa.$$

At most one of the four numbers in X is at most κ, because $X \geq 2.004$. Let ij be one of the four pairs, and kl the opposite pair. If $a_{kl} > \kappa$, then (6.4) gives $u < 1.002 - a_{ij}$.

Suppose that all four numbers are larger than κ. Then $u < 1.002 - m$, where m is the largest of them. So $S+F$ is less than the sum of the three smallest of them plus 1.005. As $S+F \geq 3.99$, each of the three is larger than 0.985. So all four pairs are full.

Suppose that one of the four numbers is at most κ, say a_{24}. Then $u < 1.002 - \max\{a_{14}, a_{23}\}$, and $S + F < a_{13} + \min\{a_{14}, a_{23}\} + 1.002 + 4\kappa \leq 3.006$. This contradicts $S + F \geq 3.99$.

Case 2: all cross sums are less than 2.004. Their sum is $2S$, so $S < 3.006$. Hence $F > 0.984$ and $S \geq 2.99$, and the sum of two opposite numbers is more than $2.99 - 2.004 = 0.986$.

Suppose that all six numbers are larger than κ. For a pairing $\alpha = ij|kl$, (6.4) gives $\max\{a_{ij}, a_{kl}\} + F - u_\alpha < 1.002$, and the maximum is at least 0.493. So $F - u_\alpha < 0.509$. Adding this for the three pairings gives $2F < 1.527$, a contradiction.

So one number is at most κ, say a_{34}. Then $a_{12} > 0.985$, so 12 is full. By (6.2), $a_{13} < 0.002$ or $a_{23} \le \kappa$; also $a_{14} < 0.002$ or $a_{24} \le \kappa$. If a number is less than 0.002, its opposite is more than 0.984, hence full. If $a_{13} < 0.002$, then 24 is full, so $a_{14} < 0.002$, and 23 is full. Then the pairs $12, 23, 24$ are full. If $a_{23} < 0.002$, then 14 is full, so $a_{24} < 0.002$, and 13 is full. Then the pairs $12, 13, 14$ are full. ∎

Lemma 6.7 (Full star). Let Y be a 4-pole in which the pairs 12, 13 and 14 are full. Then $u_\alpha(Y) \le \frac{1}{2}$ for every pairing α.

Proof. Let $\alpha = 1j|kl$, and let h be the rank of Y. Consider a path Q from 1 to j and a disjoint path Q' from k to l. The pair $1j$ is full, so every cycle of Y meets Q (Lemma 5.1). So $Y - V(Q)$ is a forest, and Q determines Q'. Hence $f_{1j|kl}(Y)$ is at most the number of paths from 1 to j in $Y - k$.

The port k lies on a cycle of Y. Otherwise both edges at k are bridges, and one of them has a side that contains neither k nor the port 1. This side contains a cycle, by the count of edge ends in the proof of Lemma 5.2, and no path from 1 to k meets it. This contradicts Lemma 5.1 for the full pair $1k$. So $Y - k$ is connected of rank $h - 1$, and it has at most 2^{h-1} paths from 1 to j by (5.1). ∎

6.4 Cuts of size four

IDEA | outline of the proof of Theorem 6.8

Let Y be the larger side of a cut of size four. In units of 2^{h+k}, the number of cycles across the cut is at most $S(Y)+F(Y)-\min_\alpha u_\alpha(Y)$, and $c(G) \ge 2^{r-1}$ requires the value 4. By the cap of Lemma 6.5, this number and the proportion of cycles of Y are together at most 4.001. So Y has few cycles, and $S(Y)+F(Y) \ge 3.997$. By Lemma 6.6 there are two possible shapes. Three full pairs at one port are impossible by Lemma 6.7. If the four pairs between two pairs of ports are full in both sides, then both sides are chains of tiles, and G is a ring.

Theorem 6.8. There is a constant r_4 with the following property. Let G be a connected cubic multigraph without loops, of rank $r \ge r_4$, that has a cyclic cut of size four and no cyclic cut of smaller size. Then G is a ring, or $c(G) < 2^{r-1}$.

Proof. Let δX be a cyclic cut of size four. By the remark at the start of the section, G has no cut of size at most two, and every cut of size three has a single vertex on one side. So $G[X]$ and $G - X$ are connected. No vertex v is on two edges of δX: otherwise the cut around $X \setminus \{v\}$, or around $X \cup \{v\}$, would be a cyclic cut of size three. So the two sides are 4-poles.

Neither side has a bridge. We show this for $G[X]$. Let e be a bridge of $G[X]$, and let A and B be the vertex sets of the two components of $G[X] - e$. In $G[A]$ every vertex has degree at

least two, except the end of e, which has degree at least one. A forest with an edge has two vertices of degree one, so $G[A]$ contains a cycle. The same holds for $G[B]$. Each edge of δX is in exactly one of the cuts δA and δB, and e is in both. So $|\delta A| + |\delta B| = 6$, and one of them is a cyclic cut of size at most three.

Call the sides X and Y, with ranks $h \le k$, and number the ports so that Lemma 6.4 applies. Then $r = h+k+3$, and k is large when r is large. We write κ for $\kappa(k)$ and assume that $k \ge h_1$, with h_1 as in Lemma 6.5. Suppose that $c(G) \ge 2^{r-1} = 4 \cdot 2^{h+k}$. We show that G is a ring.

The larger side. Put $z = 2^{-k} c(Y)$. By Lemma 6.4, and because $c(X) < 2^h$ and $h \ge 1$,

$$4 \;\le\; \frac{c(G)}{2^{h+k}} \;=\; \frac{c(X)}{2^{h+k}} + \frac{z}{2^h} + \mathcal{B} \;<\; 2^{-k} + \frac{z}{2} + \mathcal{B}. \tag{6.5}$$

By Lemma 6.4 and Lemma 6.5, $z + \mathcal{B} \le 4.001$. With (6.5) this gives $\frac{z}{2} < 2^{-k} + 0.001$, so $z \le 0.004$ and $\mathcal{B} > 3.997$ when $k \ge 10$. Hence $S(Y) + F(Y) \ge \mathcal{B} > 3.997$, and Lemma 6.6 applies to Y. It gives two cases, up to the numbering of the ports.

Case 1: the pairs $12, 13, 14$ are full in Y. By (6.2), the other three numbers $a_{ij}(Y)$ are at most κ. If one of the pairs $12, 13, 14$ is not full in X, then $a_{ij}(X) \le \frac{15}{16}$ for this pair by Theorem 5.8, and $\mathcal{B} \le 3 - \frac{1}{16} + 3\kappa + F(X)F(Y) < 3.95$. This is not possible. So the three pairs are full in X as well. By Lemma 6.7, all numbers $u_\alpha(X)$ and $u_\alpha(Y)$ are at most $\frac{1}{2}$. Let v be the largest $u_\alpha(Y)$. The numbers $u_\alpha(X)$ are at most $\frac{1}{2}$ and add up to at most 1, so

$$\sum_{\alpha \ne \beta} u_\alpha(X)\, u_\beta(Y) = \sum_{\alpha} u_\alpha(X)\big(F(Y) - u_\alpha(Y)\big) \;\le\; \frac{F(Y) + v}{2} \;\le\; \frac{3}{4}.$$

Hence $\mathcal{B} \le 3 + 3\kappa + \frac{3}{4} < 3.997$. So this case does not occur.

Case 2: the pairs $13, 14, 23, 24$ are full in Y. By (6.2), (6.3) and (6.4), $a_{12}(Y)$ and $a_{34}(Y)$ are at most κ, and $F(Y) \le 0.002 + \kappa$. If one of the four pairs is not full in X, then $\mathcal{B} \le 4 - \frac{1}{16} + 2\kappa + F(Y) < 3.95$. This is not possible. So the four pairs are full in X and in Y. Both poles have no bridge, so they are chains of tiles, from the pair $1, 2$ to the pair $3, 4$, by Corollary 5.9. The edges of the cut join the first tile of X to the first tile of Y and the last tile of X to the last tile of Y. So G is a ring. ∎

7 Girth five outside a small set

Recall that $q(G, P)$ is the probability that $\xi \cap E(G - P)$ is a forest.

Theorem 7.1. There is an integer r_5 with the following property. Let G be a connected cubic multigraph of rank $r \ge r_5$. Let P be a nonempty set of at most $3 \log r$ vertices such that $G[P]$ is connected. If $G - P$ has girth at least five, then $q(G, P) < \frac{2}{5}$.

IDEA | outline of the proof

A *piece* is a large connected part of $G - P$ with a small boundary. A piece has a forest proportion of less than $\frac{3}{5}$, for every choice of the edges outside it (Section 7.3). The reason is this. A part with more forests is thin. A long thin part contains a part in which every join is a forest. Such a part contains a ladder, and a ladder has cycles of length four. Two disjoint pieces give $q(G, P) \leq (\frac{3}{5})^2 < \frac{2}{5}$.

Suppose that $q(G, P) \geq \frac{2}{5}$. Then a family of regions with a large total weight has many pairs with exactly two links, by the second moment. These pairs give a long chain of nested separators, and two pieces lie between them (Section 7.4). We use this for short cycles (Section 7.5) and for cycles of moderate length (Section 7.6).

7.1 Nested separators

Lemma 7.2 (Nested separators)**.** Let u and p be two vertices of a connected graph H. Let C_1 and C_2 be disjoint connected sets of vertices that avoid u and p, such that u and p lie in different components of $H - C_i$ for $i = 1, 2$. Let U_i be the vertex set of the component of u in $H - C_i$. Then exactly one of the following holds: $C_1 \subseteq U_2$ and $U_1 \subseteq U_2$; or $C_2 \subseteq U_1$ and $U_2 \subseteq U_1$.

Proof. The set C_2 is connected and disjoint from C_1, so it lies in one component of $H - C_1$. Suppose that this component is not U_1. Then U_1 avoids C_2. Some edge joins U_1 to C_1, because H is connected. So $U_1 \cup C_1$ is a connected set that contains u and avoids C_2, and therefore $U_1 \cup C_1 \subseteq U_2$. This is the first alternative. In the same way, the second alternative holds if C_1 does not lie in U_2.

Suppose that $C_2 \subseteq U_1$ and $C_1 \subseteq U_2$. Let W_i be the vertex set of the component of p in $H - C_i$. Then W_2 avoids C_1, and the argument above, with p in the place of u, gives $W_2 \cup C_2 \subseteq W_1$. This is impossible, because $C_2 \subseteq U_1$, and U_1 is disjoint from W_1. The two alternatives exclude each other, because $C_1 \subseteq U_2$ and $U_2 \subseteq U_1$ would give $C_1 \subseteq U_1$. ■

So a family of pairwise disjoint connected sets that separate u from p is linearly ordered: write $C_1 < C_2$ in the first case of the lemma. The order is transitive, because it is the order of the sets U_i by inclusion.

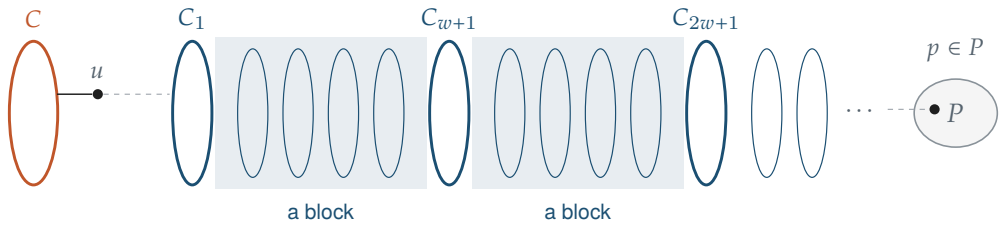


Figure 8. Nested separators. The vertex u has a neighbour in a set C, and each set C_i separates u from p in the graph without C. So the sets C_i are nested (Lemma 7.2). Every w-th set is the end of a block, and the edges that leave a block end in its two end sets (Lemma 7.3).

Lemma 7.3 (Blocks). Let u and p be two vertices of a connected graph H. Let $C_1 < C_2 < \cdots < C_n$ be pairwise disjoint connected sets that avoid u and p and separate u from p, and let $w \geq 1$ be an integer. Then there are pairwise disjoint sets $Y_l \subseteq V(H)$, one for each integer $l \geq 0$ with $(l+1)w \leq n-1$, such that

(i) Y_l contains $w-1$ of the sets C_i;

(ii) every edge of H that leaves Y_l ends in C_{lw+1} or in $C_{(l+1)w+1}$;

(iii) Y_l lies in the component of u in $H - C_{(l+1)w+1}$.

Proof. Let U_i be the vertex set of the component of u in $H - C_i$. Then $U_i \cup C_i \subseteq U_j$ for $i < j$. For $i < j$ put $X_{ij} = U_j \setminus (U_i \cup C_i)$. An edge of H that leaves U_j ends in C_j, because U_j is a component of $H - C_j$. No edge joins X_{ij} to U_i: its end in X_{ij} would lie in the component U_i of $H - C_i$. So every edge of H that leaves X_{ij} ends in C_i or in C_j. If $i < l < j$, then $C_l \subseteq U_j$, and C_l avoids $U_i \cup C_i \subseteq U_l$; so $C_l \subseteq X_{ij}$. If $i < j \leq i' < j'$, then $X_{ij} \subseteq U_j \subseteq U_{i'}$ is disjoint from $X_{i'j'}$. So the sets $Y_l = X_{lw+1,(l+1)w+1}$ have the three properties. ■

7.2 Joins

Let N be a connected subcubic multigraph. Its vertices of degree less than three are its *boundary vertices*, and its *deficit* $d(N)$ is the sum of $3 - \deg v$ over all vertices v. A count of edge ends gives

$$|V(N)| = 2\beta(N) - 2 + d(N). \tag{7.1}$$

Let T be a set of boundary vertices of even size. A *T-join* of N is a set of edges in which the vertices of T have odd degree and all other vertices have even degree. There are $2^{\beta(N)}$ of them. The *forest proportion* of T is the proportion of the T-joins that are forests, and T is *rigid* if every T-join is a forest. A region A of N is *T-free* if it contains no vertex of T and every component of $N - A$ contains an even number of vertices of T.

Lemma 7.4 (Free regions). Let J be a uniformly random T-join of N. For a region A let E_A be the event that $J \cap \delta A = \emptyset$ and $J \cap E(N[A]) \neq \emptyset$.

(i) Let A be a region without a vertex of T. Some T-join avoids δA if and only if A is T-free.

(ii) If A is T-free, then E_A has the same probability as for a uniformly random even subgraph of N.

(iii) If A and B are disjoint T-free regions, then $E_A \cap E_B$ is empty, or it has the same probability as for a uniformly random even subgraph of N.

So the bound of Lemma 2.4 holds for J and pairwise disjoint T-free cyclic regions: if there are n of them, each with at most d boundary edges, then the forest proportion of T is at most $\frac{1}{2} + (d+1)2^{d-1}/n$.

Proof. (i) A T-join that avoids δA has an even number of vertices of odd degree in each component of $N - A$. In the other direction, each component K of $N - A$ has a $(T \cap K)$-join if $|T \cap K|$ is even. The union of these joins is a T-join F without an edge that has an end in A.

(ii) The map $J \mapsto J + F$ is a bijection from the T-joins to the even subgraphs of N, and it keeps the edges that have an end in A. The event E_A depends on these edges only.

(iii) Suppose that some T-join is in $E_A \cap E_B$. Delete its edges in $N[A]$ and in $N[B]$. These form even subgraphs, so the result is a T-join F' without an edge that has an end in $A \cup B$. Use the map $J \mapsto J + F'$.

The proof of Lemma 2.4 uses the probabilities of the events E_i and upper bounds for the probabilities of the events $E_i \cap E_j$. By (ii) and (iii) these hold for J. If J is a forest, then no event E_i occurs. ■

Lemma 7.5 (Product). Let N be a multigraph, and let P and T be sets of vertices such that N has a T-join. Let $U_1, \dots, U_m \subseteq V(N) \setminus (P \cup T)$ be pairwise disjoint sets of vertices of degree three such that each $N[U_i]$ is connected. Let $f_1, \dots, f_m$ be numbers such that in $N[U_i]$ every even set of boundary vertices has forest proportion at most f_i. Then $J \cap E(N - P)$ is a forest for at most the proportion $f_1 \cdots f_m$ of the T-joins J of N. For a cubic multigraph G and $T = \emptyset$ this says that $q(G, P) \le f_1 \cdots f_m$.

Proof. Let E' consist of the edges that lie in none of the graphs $N[U_i]$, and let J be a uniformly random T-join. Fix a set $x \subseteq E'$ that extends to a T-join. For each i, let T_i be the set of the vertices of U_i that are incident with an odd number of edges of x. The set U_i avoids T, so the parity equations say that $J \cap E(N[U_i])$ is a T_i-join. These equations involve disjoint sets of edges, so the joins are independent and uniform, conditional on $J \cap E' = x$. A vertex of degree three in $N[U_i]$ has no incident edge in E'. So T_i is an even set of boundary vertices. If $J \cap E(N - P)$ is a forest, its restriction to every $N[U_i]$ is a forest. The conditional probability is therefore at most $f_1 \cdots f_m$. Average over x. ■

The next two lemmas say that a multigraph of girth five with a rigid set is small. The ladder shows that this is false for girth four: the two ends of one rail form a rigid set.

Lemma 7.6 (Full pairs at girth five). Let M be a connected subcubic multigraph of girth at least five. Let $x \ne y$ be vertices of degree at most two such that every $\{x, y\}$-join of M is a path. Then $|V(M)| < 15\, d(M)$.

Proof. Put $d = d(M)$ and $b = \beta(M)$. If $b = 0$, then M is a tree and $|V(M)| = d - 2$. Let $b \ge 1$. Delete a vertex of degree one other than x and y, as long as there is one. If then x has degree one, delete x and take its neighbour in the place of x; do the same for y; repeat. These steps keep the rank, and every join is a path after each step. The two vertices stay

distinct, because $b \geq 1$. The result M_1 is a pole in which the two vertices are ports and form a full pair. A port of M_1 has degree two in M, or one of its edges in M leads to a deleted tree, and such a tree has a deficit of at least two. So M_1 has at most d ports.

By Lemma 5.2, replace the other ports by edges. The result M_1° consists of bridges and of bridgeless 2-poles $N_1, \dots, N_q$ with full pairs of ports, and their ranks $b_i \geq 1$ add up to b. An edge of M_1° is a path of M_1 whose inner vertices are ports; call it *plain* if this path has no inner vertex. At most $d - 2$ edges are not plain. A cycle of M_1° with at most four edges is not a cycle of M, so one of its edges is not plain. By Lemma 5.6, N_i is a pair of parallel edges, or a capped ladder with some number m of tiles. In the second case its rank is at most $2m + 1$, and it contains at least $m/3$ pairwise edge-disjoint cycles with at most four edges: the squares, pairs of consecutive single rungs with the interface between them, and one cycle through a cap. So in both cases at least $b_i/7$ edges of N_i are not plain. Hence $b \leq 7(d - 2)$, and (7.1) gives $|V(M)| < 15d$. ■

Lemma 7.7 (Rigid sets). Let M be a connected subcubic multigraph of girth at least five with deficit d. Let $\lambda : V(M) \to \mathbb{Z}$ be a map with $|\lambda(u) - \lambda(v)| \leq 1$ for every edge uv, such that fewer than W vertices have any given value. If some even set U of boundary vertices is rigid, then $|V(M)| \leq 500\, dW(W + d)$.

Proof. If $U = \emptyset$, then M has no cycle, and $|V(M)| = d - 2$. Let $U \neq \emptyset$. Let S be a connected subgraph of M that contains U and has as few edges as possible. It is a tree, and its vertices of degree one are in U. Call a vertex of S *special* if it is in U or has degree three in S. There are at most $2|U| - 2$ special vertices. A *segment* is a path of S between two special vertices without a special inner vertex. The tree S is the union of at most $2|U| - 3$ segments.

Put $\rho = 30W + 8d$ and $\ell_0 = (2\rho + 1)W + 2$. Suppose that some segment σ has at least $2\ell_0$ edges, and let z_0 be a vertex of σ with at least ℓ_0 edges of σ on each side. Let B be the vertex set of the component of z_0 in the subgraph of M induced by the vertices v with $|\lambda(v) - \lambda(z_0)| \leq \rho$. Then $M[B]$ is connected, B has fewer than $(2\rho + 1)W$ vertices, and it contains the $2\rho + 1$ vertices of σ nearest to z_0.

The tree meets B in one part of σ. Let v be a vertex of S in B, and let π be a path from v to z_0 in $M[B]$. It has fewer than $\ell_0 - 2$ edges. If v is not an inner vertex of σ, or if σ has more edges between v and z_0 than π has, then π can replace a part of σ with more edges, and S does not have the fewest edges. So the vertices of S in B are inner vertices of σ, and B contains no vertex of U. Let S^- and S^+ be the components of $S - B$ that contain the two ends of σ. They are different, and every vertex of U is in one of them. Let x^- be the first vertex of σ in B, seen from the end in S^-, and let x^+ be the first one seen from the other end. Then $x^- \neq x^+$, the vertex x^- has a neighbour in S^-, and x^+ has a neighbour in S^+. Let K^- and K^+ be the components of $M - B$ that contain S^- and S^+, and put $U^- = U \cap V(S^-)$.

The set B is small. We claim that $|B| < 15\, d(M[B])$. Suppose first that $K^- = K^+$ or that $|U^-|$ is even. Then $M - B$ contains a U-join. Its union with a cycle of $M[B]$ would be a U-join that is not a forest. So $M[B]$ is a tree, and $|B| = d(M[B]) - 2$. Suppose now that $K^- \neq K^+$ and

that $|U^-|$ is odd. Let J be an $\{x^-, x^+\}$-join of $M[B]$. Add an edge from x^- to a neighbour v^- in K^-, an edge from x^+ to a neighbour v^+ in K^+, a join of K^- for the set U^- with v^- added or removed, and a join of K^+ for the set $U \setminus U^-$ with v^+ added or removed. The result is a U-join of M. So J is a forest, and hence a path. The vertices x^- and x^+ have degree at most two in $M[B]$, and the claim is Lemma 7.6.

An edge of M that leaves B starts in a vertex v with $|\lambda(v) - \lambda(z_0)| = \rho$. There are fewer than $2W$ such vertices, and each has at most two such edges. So $d(M[B]) < 4W + d$, and the claim gives $|B| < 60W + 15d < 2\rho + 1$. This is a contradiction. So every segment has fewer than $2\ell_0$ edges.

The count. So S has fewer than $4|U|\,\ell_0 \le 4d\,\ell_0$ vertices. The set $V(S)$ is connected and contains U. So $M - V(S)$ is a forest, as in the first case above. Let it have f vertices. At most $|V(S)| + 2$ edges join it to S, and a count of edge ends in the forest gives $f \le |V(S)| + d$. Hence $|V(M)| \le 8d\,\ell_0 + d \le 500\,dW(W + d)$. ∎

7.3 Pieces with a small boundary

Put

$$R_* = R_{1/20}, \qquad B = 2R_* + 2^{100} + 1,$$

with $R_{1/20}$ as in Section 4. Then $\Theta_{1/20}(k) + k \le B(k+1)^2$ for all $k \ge 1$.

Lemma 7.8 (Pieces). There is an absolute constant $C_1 \ge 2$ with the following property. Let $k \ge 1$, and let N be a connected subcubic multigraph of girth at least five with at most k boundary vertices and at least $C_1 k^7$ vertices. Then every set of boundary vertices of even size has forest proportion less than $\frac{3}{5}$.

Proof. For $k \ge 1$ put $W_k = B(k+1)^2$, and let s_k be the least integer with $2^{s_k} > (2s_k + 1)W_k$. Then $2^{s_k} \le 2(2s_k + 1)W_k$ and $s_k = O(\log k)$. Fix k_0 such that $4s_k < k$ for all $k \ge k_0$. We define numbers $F(k) \ge 2k$ below. Let N have at most k boundary vertices, and let T be an even set of boundary vertices with forest proportion at least $\frac{3}{5}$. We show by induction on k that N has fewer than $F(k)$ vertices. If T is empty, then N has no cycle, and it has fewer than $2k$ vertices. So let T be not empty. Fix a boundary vertex o, and call the set of the vertices at distance j from o the *layer* j. Put $W = W_k$ and $s = s_k$.

The layers are small. Take a breadth-first tree of N from o, and let P be the union of its paths from o to the other boundary vertices. Then $N[P]$ is connected, every vertex outside P has degree three, and P has at most k vertices in each layer. The tree paths inside P contain a T-join F_0. The map $J \mapsto J + F_0$ is a bijection from the T-joins of N to the even subgraphs of N, and $J + F_0$ has the same edges in $N - P$ as J. So if J is a forest, then $J + F_0$ is a forest in $N - P$. Hence $q(N, P) \ge \frac{3}{5} > \frac{1}{2} + \frac{1}{20}$. The distance from o is a level function. By Lemma 4.1 with $\delta = \frac{1}{20}$, each layer has fewer than $\Theta_{1/20}(k)$ vertices outside P. So each layer has fewer than W vertices.

Few boundary vertices. Let $k < k_0$. Put $h = 10^4\,W^3$ and $F(k) = Wh\,(k + 2^{Wh} + 2)$, and suppose

that N has at least $F(k)$ vertices. Then it has at least $h\,(k+2^{Wh}+2)$ layers. Divide the layers $1, 2, \dots$ into groups of h consecutive layers. At least 2^{Wh} groups contain no boundary vertex. In such a group let M be a component of the subgraph induced by the group that has a vertex v in the last layer of the group. A shortest path from o to v shows that M has a vertex in each layer of the group, so $|V(M)| \geq h$. The vertices of M have degree three in N, and an edge that leaves M starts in the first or in the last layer of the group. So $d(M) < 4W$, and the rank of M is less than Wh. By Lemma 7.7, with the distance from o in the place of λ, a rigid set of M would give $|V(M)| < 500 \cdot 4W \cdot W \cdot 5W = h$. So no even set of boundary vertices of M is rigid, and each of them has forest proportion at most $1 - 2^{-Wh}$. By Lemma 7.5, the forest proportion of T is at most $(1-2^{-Wh})^{2^{Wh}} < \frac{1}{2}$. This is a contradiction.

Many boundary vertices. Let $k \geq k_0$, and put $k' = 4s$. Then $k' < k$. Put

$$w = \lceil k'F(k')/5 \rceil + 1, \qquad n_1 = 5(2s+1)\,2^{2s}, \qquad n_2 = k^2(k+2)\,w,$$
$$F(k) = 3 \cdot 2^s \big(k + 2s\,(k + n_1 + n_2 + 2)\big) + 1,$$

and suppose that N has at least $F(k)$ vertices. Call a cycle *short* if its length is at most $2s$. Let $\mathcal{D}$ be a maximal family of pairwise disjoint short cycles, and let t be their number. Let w_0 be a vertex, and let W_0 be the set of the vertices within distance s of w_0. Suppose that W_0 contains no boundary vertex and no vertex of a cycle of $\mathcal{D}$. Take a breadth-first tree from w_0. If some vertex at distance less than s from w_0 had fewer than two children in this tree, or w_0 fewer than three, then $N[W_0]$ would contain a short cycle, and this cycle would meet a cycle of $\mathcal{D}$. So $|W_0| \geq 2^s$. But W_0 meets at most $2s+1$ layers, and $2^s/(2s+1) > W$. This contradicts the first step. A ball of radius s has at most $3 \cdot 2^s$ vertices. Hence N has at most $3 \cdot 2^s(k + 2s\,t)$ vertices, and $t \geq k + n_1 + n_2 + 2$.

At most k cycles of $\mathcal{D}$ contain a boundary vertex. Each of the other cycles Q has at most $2s$ boundary edges, and $V(Q)$ contains no vertex of T. Call Q *free* if $V(Q)$ is T-free. So more than n_1 cycles are free, or more than n_2 cycles are not free. In the first case Lemma 7.4 with $d = 2s$ shows that the forest proportion of T is less than $\frac{1}{2} + (2s+1)\,2^{2s-1}/n_1 = \frac{3}{5}$. This is a contradiction.

In the second case let Q be a cycle that is not free. Some component of $N - V(Q)$ contains an odd number of vertices of T. The set T is even and avoids Q. So T has a vertex a in this component and a vertex b in another component, and $V(Q)$ separates a from b. There are fewer than k^2 pairs a, b. So more than $(k+2)w$ of these cycles separate the same two vertices a and b. By Lemma 7.2 their vertex sets are linearly ordered, and Lemma 7.3 for N, a and b gives $k+2$ pairwise disjoint sets Y_l. At most k of them contain a boundary vertex of N. Let Y_1 and Y_2 be two of the others, and let $i \in \{1, 2\}$. The set Y_i contains $w-1$ of the cycles, and each cycle has at least five vertices. So $|Y_i| \geq 5(w-1) \geq k'F(k')$. At most $2 \cdot 2s = k'$ edges leave Y_i, and every component of $N[Y_i]$ has such an edge, because $b \notin Y_i$. So some component has a vertex set Y_i' with at least $F(k')$ vertices. The vertices of Y_i have degree three in N, so $N[Y_i']$ has at most k' boundary vertices. By induction, every even set of boundary vertices of $N[Y_i']$ has forest proportion less than $\frac{3}{5}$. Lemma 7.5, for $N, T, P = \emptyset$ and the disjoint sets Y_1' and Y_2', shows that the forest proportion of T is at most $\frac{9}{25}$. This is a contradiction.

The size of F. We have $2^s = O(k^2 \log k)$ and $s = O(\log k)$. So $n_1 = O(k^4 \log^3 k)$ and $n_2 = O\big(k^3 \log k \cdot F(4s_k)\big)$, and the definition gives $F(k) = O\big(k^6 \log^5 k + k^5 \log^3 k \cdot F(4s_k)\big)$ for $k \geq k_0$. Hence $F(k) \leq C_1 k^7$ for an absolute constant C_1, by induction on k. ■

7.4 Heavy families

Let C_1 be as in Lemma 7.8.

IDEA | two links give a separator

If two disjoint connected sets have exactly two links, then one of them separates a neighbour of the other one from a fixed vertex p. Disjoint sets that separate the same two vertices are nested, and the parts between them have small boundaries (Figure 8). So a long chain of such sets contains two pieces.

Proposition 7.9 (A long chain). Let G be a connected cubic multigraph, and let P be a nonempty set of vertices such that $G[P]$ is connected and $G - P$ has girth at least five. Let $p \in P$, let A be a region that avoids P, and let $u \notin A$ be a vertex in the component of p in $G - A$. Let $C_1, \dots, C_T$ be pairwise disjoint regions that avoid P, A and u and separate u from p in $G - A$. Let $\Lambda \geq 1$ be an integer and $m \geq 1$ a number such that A and every C_i have at most Λ boundary edges, and every C_i has at least m vertices. If

$$T > (\Lambda + 2)\big(\lceil 2^8 C_1 \Lambda^8 / m \rceil + 1\big),$$

then $q(G, P) \leq \frac{9}{25}$.

Proof. Each C_i is connected and separates u from p in $G - A$. So it lies in the component H of $G - A$ that contains u and p. By Lemma 7.2 we may assume that $C_1 < C_2 < \cdots < C_T$. Put $w = \lceil 2^8 C_1 \Lambda^8 / m \rceil + 1$. Lemma 7.3 gives at least $\Lambda + 2$ pairwise disjoint sets Y_l. The set P is connected, it contains p, and it is disjoint from A and from the sets C_i. So P lies in the component of p in $H - C_i$ for every i, and every Y_l avoids P. At most Λ of the sets Y_l contain a vertex with a neighbour in A. Let Y be one of the others. The edges of G that leave Y are edges of H, so $|\delta Y| \leq 2\Lambda$, and $|Y| \geq (w-1)m \geq 2\Lambda \cdot C_1 (2\Lambda)^7$. Every component of $G[Y]$ has an edge that leaves Y. So there are at most 2Λ components, and one of them has a vertex set Y' with at least $C_1(2\Lambda)^7$ vertices. A boundary vertex of $G[Y']$ is an end of an edge of δY, so there are at most 2Λ of them, and $G[Y']$ has girth at least five. By Lemma 7.8, every even set of boundary vertices of $G[Y']$ has forest proportion less than $\frac{3}{5}$. There are two sets Y, and Lemma 7.5 for their two sets Y' gives $q(G, P) \leq \frac{9}{25}$. ■

Lemma 7.10 (Two links)**.** Let G be a connected multigraph, and let p be a vertex. Let $\mathcal{A}$ be a finite family of regions that avoid p, with weights $w_A \geq 0$ of total $\mu > 0$. Let W_2 be the sum of $w_A w_{A'}$ over the ordered pairs (A, A') of disjoint regions of $\mathcal{A}$ with $\ell(A, A') = 2$. Let N and e be numbers such that, for every $A \in \mathcal{A}$, at most N vertices outside A have a neighbour in A, and the regions of $\mathcal{A}$ that are disjoint from A and joined to A by an edge have total weight at most e. Let K be a number with $W_2 \geq \frac{2}{3}\mu^2 - K\mu$ and $\mu > \frac{3}{2}K + 3e$. Then there are a region $A \in \mathcal{A}$ and a vertex $u \notin A$ in the component of p in $G - A$ such that the regions of $\mathcal{A}$ that are disjoint from A, avoid u, and separate u from p in $G - A$ have total weight at least $(\mu/3 - K/2 - e)/N$.

Proof. Let $A, A' \in \mathcal{A}$ be disjoint. Then A' lies in the component of p in $G - A$, or A lies in the component of p in $G - A'$. Indeed, suppose that the first statement is false. A path from p to its first vertex in A stays in the component of p in $G - A$ before it ends, so it avoids A'. As A is connected and disjoint from A', all of A lies in the component of p in $G - A'$.

Call an ordered pair (A, A') *good* if the regions are disjoint, $\ell(A, A') = 2$, no edge joins them, and A' lies in the component of p in $G - A$. By the last paragraph, the pairs with the first, the second and the fourth property have total weight at least $W_2/2$. The pairs that are joined by an edge have total weight at most $e\mu$. So the good pairs have total weight at least $W_2/2 - e\mu \geq \mu(\mu/3 - K/2 - e)$, and for some $A \in \mathcal{A}$ the regions A' with (A, A') good have total weight at least $\mu/3 - K/2 - e > 0$. Fix this A.

Let (A, A') be good. No edge joins the two regions, so the two links are components Y_1, Y_2 of $G - A - A'$ that are joined to both. The vertex p lies in at most one of them. Let Y be one that avoids p, and let u be a vertex of Y with a neighbour in A. The set Y is connected and joined to A', and A' lies in the component of p in $G - A$. So u lies in that component as well. And Y is a component of $G - A - A'$ that contains u and not p. So A' avoids u and separates u from p in $G - A$. There are at most N vertices outside A with a neighbour in A. For one of them, u, the regions A' have the total weight of the lemma. ∎

Proposition 7.11 (Heavy families). Let G be a connected cubic multigraph, and let P be a nonempty set of vertices such that $G[P]$ is connected, $G - P$ has girth at least five, and $q(G,P) \geq \frac{2}{5}$. Let $\mathcal{A}$ be a finite family of regions that avoid P, with weights $w_A \geq 0$ of total μ, and let W_2 be as in Lemma 7.10. Let $\Lambda \geq 1$ be an integer, and let $m \geq 1$, $\kappa > 0$ and $K \geq 0$ be numbers such that

(a) every region of $\mathcal{A}$ has at most Λ boundary edges and at least m vertices;

(b) for every $A \in \mathcal{A}$, the regions of $\mathcal{A}$ that meet A have total weight at most κ;

(c) some random variable $Z \geq 0$ has $\mathbb{E}Z = \mu$ and $\mathbb{E}Z^2 \leq \mu^2 + W_2 + K\mu$, and $Z = 0$ whenever $\xi \cap E(G - P)$ is a forest.

Then

$$\mu \leq \tfrac{3}{2}K + 3\Lambda\kappa\,(\Lambda+3)\bigl(\lceil 2^8 C_1 \Lambda^8/m\rceil + 1\bigr).$$

Proof. Put $w = \lceil 2^8 C_1 \Lambda^8/m\rceil + 1$. We may assume that $\mu > \frac{3}{2}K + 3\Lambda\kappa$. By (c), $\mathbb{P}(Z \neq 0) \leq 1 - q(G,P) \leq \frac{3}{5}$, and the Cauchy–Schwarz inequality $\mu^2 \leq \mathbb{E}Z^2 \cdot \mathbb{P}(Z \neq 0)$ gives $W_2 \geq \frac{2}{3}\mu^2 - K\mu$.

Fix $p \in P$. We apply Lemma 7.10. Let $A \in \mathcal{A}$. By (a), at most Λ vertices outside A have a neighbour in A, and a region that is joined to A by an edge contains one of them. The regions of $\mathcal{A}$ that contain a given vertex all meet one of them, so their total weight is at most κ by (b). Hence we may take $N = \Lambda$ and $e = \Lambda\kappa$. The lemma gives a region $A \in \mathcal{A}$, a vertex $u \notin A$ in the component of p in $G - A$, and regions of $\mathcal{A}$ of total weight at least $(\mu/3 - K/2 - \Lambda\kappa)/\Lambda$ that are disjoint from A, avoid u, and separate u from p in $G - A$. Take a maximal family $C_1, \dots, C_T$ of pairwise disjoint regions among them. Each of these regions meets some C_i, so their total weight is at most $T\kappa$ by (b). As $q(G,P) > \frac{9}{25}$, Proposition 7.9 gives $T \leq (\Lambda+2)\,w$. So $\mu/3 - K/2 - \Lambda\kappa \leq \Lambda\kappa\,(\Lambda+2)\,w$, and the claim follows because $w \geq 1$. ∎

7.5 Packing of short cycles

IDEA | the purpose of the packing bound

The second-moment argument requires a graph of girth about $\log r$. So we delete a maximal family of disjoint short cycles, and we need a bound for the size of this family. Lemma 2.4 gives such a bound only when the proportion of forests exceeds one half by a fixed margin. Here the bound follows from Proposition 7.11.

Proposition 7.12 (Packing). Let G be a connected cubic multigraph, and let P be a nonempty set of vertices such that $G[P]$ is connected. Suppose that $q(G,P) \geq \frac{2}{5}$ and that $G - P$ has girth at least five. Then, for every integer $d \geq 5$, the multigraph $G - P$ has at most $2^{10} C_1\, 2^d d^{10}$ pairwise disjoint cycles of length at most d.

Proof. Let $Q_1, \dots, Q_n$ be such cycles, and put $X_i = V(Q_i)$. Every vertex of Q_i lies on exactly one edge outside Q_i, and Q_i has at least five vertices. So X_i is a cyclic region with at most d boundary edges and at least five vertices. Let E_i be the event that O_{X_i} holds and $\xi \cap E(G[X_i]) \neq \emptyset$. Put $p_i = \mathbb{P}(E_i)$, let Z be the number of the events E_i that occur, and put $\mu = \mathbb{E}Z$. As in the proof of Lemma 2.4, we have $p_i \geq 2^{-d-1}$ and $\mathbb{P}(E_i \cap E_j) = 2^{\ell(X_i,X_j)-1} p_i p_j$ for $i \neq j$, and for each i at most d indices j have $\ell(X_i, X_j) \geq 3$. Let W_2 be the sum of $p_i p_j$ over the ordered pairs (i, j) with $\ell(X_i, X_j) = 2$. Then $\mathbb{E}Z^2 \leq \mu^2 + W_2 + (d+1)\mu$. If $\xi \cap E(G - P)$ is a forest, then no event E_i occurs, and $Z = 0$.

So Proposition 7.11 applies to the sets X_i with the weights p_i, and with $\Lambda = d$, $m = 5$, $\kappa = 1$ and $K = d + 1$; property (b) holds because the sets are pairwise disjoint and $p_i \leq 1$. It gives

$$\mu \;\leq\; \tfrac{3}{2}(d+1) + 3d\,(d+3)\big(\lceil 2^8 C_1 d^8/5\rceil + 1\big) \;\leq\; 2^9 C_1\, d^{10},$$

and $n \leq 2^{d+1}\mu$. ∎

7.6 Proof of Theorem 7.1

Let G and P be as in the theorem, and suppose that $q(G, P) \geq \frac{2}{5}$. We derive a contradiction when r is large. All limits below are for $r \to \infty$, and they are uniform in G and P. Put

$$x = \log r, \qquad A = 6\,e^{124}, \qquad L = \lfloor Ax \rfloor, \qquad D = \lceil x/16 \rceil.$$

Cycles of moderate length. Let S be the union of the vertex sets of a maximal family of pairwise disjoint cycles of length at most D in $G - P$. By Proposition 7.12, $|S| \leq 2^{10} C_1\, 2^D D^{11} \leq r/x^2$ for large r. We use Proposition 3.5 for G, P, D, L and S. Its hypotheses hold for large r: the set P has at most $3x$ vertices and $G[P]$ is connected, $x/16 \leq D \leq x/8$, and $7x \leq L \leq x\sqrt{\log x}$. So we are in the setting of Section 3.1, with $\Gamma = G$, $U = P \cup S$ and $J = G - U$. The set $\mathcal{C}$ consists of the cycles of J of length at most L, and the numbers s, a_C, μ and W_2 and the random variable Z are defined there. With $\eta = r^{-1/60}$ the proposition gives

$$L^2 2^{-s} \leq \eta, \qquad \mu \;\geq\; \frac{1}{4}\ln\frac{L}{6x} - \frac{1}{\sqrt{x}} \;>\; 30, \qquad \mathbb{E}Z^2 \;\leq\; \mu^2 + W_2 + (1+\eta)\mu.$$

The bound $\mu > 30$ holds because $L/(6x) \geq e^{124} - 1$.

The contradiction. We apply Proposition 7.11 to the vertex sets of the cycles of $\mathcal{C}$, with the weights a_C, and with $\Lambda = L$, $m = x/16$, $\kappa = \eta$ and $K = 1 + \eta$. A cycle $C \in \mathcal{C}$ has at most L vertices, each of them has one edge end outside C, and the girth of J is greater than $D \geq x/16$. So $V(C)$ has at most L boundary edges and at least $x/16$ vertices; this is (a). A cycle of $\mathcal{C}$ that meets C passes through one of the at most L vertices of C, so (b) holds by (3.1). If $\xi \cap E(G - P)$ is a forest, then ξ contains no cycle of J, and $Z = 0$; this is (c). So

$$\mu \;\leq\; \tfrac{3}{2}(1+\eta) + 3L\eta\,(L+3)\big(\lceil 2^{12} C_1 L^8/x\rceil + 1\big) \;=\; \tfrac{3}{2} + o(1),$$

because $L \leq Ax$ and $\eta = r^{-1/60}$. This contradicts $\mu > 30$, and Theorem 7.1 is proved. ∎

8 Proof of Theorem 1.1

Let K_0 be as in Section 5. By (1.1),

$$H(r) \;\geq\; 2^{r-1} + (r-2)\,2^{r/2-1} \qquad \text{for all } r \geq 3. \tag{8.1}$$

Choose $R \geq 9$ so large that the following holds for every $r \geq R$: Proposition 6.3, Theorem 6.8 and Theorem 7.1 apply to the rank r; and $r - 2 > 10K_0^2$.

Let $r \geq R$. A ring R_r has $H(r)$ cycles by Corollary 6.2, so $\psi(r) \geq H(r)$. By Lemma 3.6, some connected cubic multigraph G of rank r has $\psi(r)$ cycles. We show that G is a ring or that $c(G) \leq 2^{r-1} + 5K_0^2\,2^{r/2}$. In the first case $c(G) \leq H(r)$ by Corollary 6.2. In the second case $c(G) < H(r)$ by (8.1), because $r - 2 > 10K_0^2$. So $\psi(r) \leq H(r)$.

Case 1: G has a loop e. Every cycle of G other than e is a cycle of $G - e$, which has rank $r - 1$. So $c(G) \leq 1 + (2^{r-1} - 1)$.

Case 2: G has no loop, and it has a cyclic cut of size at most three. Then $c(G) \leq 2^{r-1} + 5K_0^2\,2^{r/2}$ by Proposition 6.3.

Case 3: G has no loop, it has a cyclic cut of size four, and it has no smaller cyclic cut. Then G is a ring or $c(G) < 2^{r-1}$, by Theorem 6.8.

Case 4: G has no loop and no cyclic cut of size at most four. Let Z be a cycle with at most four edges. Then at most four edges leave $V(Z)$, and at most eight edges have an end in $V(Z)$. So $G - V(Z)$ has rank at least $r - 8 > 0$, and $\delta V(Z)$ is a cyclic cut of size at most four. Hence G has girth at least five. The graph G has $2r - 2 \geq 8$ vertices. Choose a set P of eight vertices such that $G[P]$ is connected. Then $|P| \leq 3\log r$, and $G - P$ has girth at least five. So $q(G, P) < \frac{2}{5}$ by Theorem 7.1.

Let Z be a cycle of G that meets P. Its edges in $G - P$ form a proper subset of Z, hence a forest. So at most $q(G, P)\,2^r < \frac{2}{5}\,2^r$ cycles meet P. The other cycles are cycles of $G - P$, and there are at most $2^{\beta(G-P)}$ of them. Let a be the number of edges of $G[P]$. Then $24 - 2a$ edges leave P, and $G - P$ arises from G by the deletion of 8 vertices and $24 - a$ edges. So

$$\beta(G - P) = r - 17 + a + \mathrm{comp}(G - P).$$

The graph G has no cut of size one, because such a cut is cyclic (Section 6). So every component of $G - P$ contains the ends of at least two of the edges that leave P. Hence $\mathrm{comp}(G - P) \leq 12 - a$ and $\beta(G - P) \leq r - 5$. Therefore $c(G) < \left(\frac{2}{5} + \frac{1}{32}\right)2^r < 2^{r-1}$. ∎

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